

A LOCAL EXTENSION CHARACTERIZATION OF OKA MANIFOLDS

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ABSTRACT. We characterize Oka manifolds by the local extension approximation property (LEAP), which permits holomorphic maps to be approximated on enlarged planar domains without losing holomorphic dependence on auxiliary variables. We prove that LEAP yields convex approximation in every source dimension and propagates across proper closed complex analytic subsets. It follows that strong dominability characterizes Oka manifolds, that convex approximation in the target dimension suffices, and that the Oka property is local in the Euclidean topology. Local families of entire curves spanning the tangent spaces off a proper analytic subset already force the Oka property. Every smooth complex projective rationally connected manifold, hence every smooth complex Fano manifold, is Oka. We also prove that the complement of a smooth degree- d hypersurface in $\mathbb{P}^n$ is Oka exactly when $d \leq n + 1$. Further applications include abelian fibrations, Schoen threefolds, and spaces of rational maps.

1. INTRODUCTION

The Oka principle concerns the existence, approximation, and interpolation of holomorphic maps from Stein spaces. Its modern formulations grew from Oka's work on the second Cousin problem and Grauert's theory of holomorphic fiber bundles over Stein spaces [45, 35]. Gromov introduced dominating holomorphic sprays and proved the Oka principle for elliptic manifolds [36]; Forstnerič extended this theory to subelliptic manifolds [20]. He established the equivalence of convex approximation with the basic Oka property [23, Theorem 0.1], including approximation and jet interpolation [22, Proposition 1.2 and Corollary 1.3], and subsequently proved that convex approximation implies the parametric Oka property [24, Theorem 1.1]. For general background, see the survey of Forstnerič and Lárusson [28] and Forstnerič's monograph [26].

The present paper characterizes this global approximation property by a local planar extension operation and applies the characterization to projective manifolds, divisor complements, and spaces of rational maps.

Definition 1.1 ([24, Definition 1.2]). A complex manifold Y is an *Oka manifold* if it has the *convex approximation property* CAP: for every $m \geq 1$, compact convex set $K \subset \mathbb{C}^m$, holomorphic map $f : U \rightarrow Y$ on an open neighborhood U of K , and $\epsilon > 0$, there is $F \in \mathcal{O}(\mathbb{C}^m, Y)$ such that

$$\sup_{z \in K} d_Y(F(z), f(z)) < \epsilon,$$

where d_Y is any distance inducing the topology of Y .

Unless explicitly stated otherwise, every complex manifold in this paper is connected. Although CAP is formulated on compact convex subsets of Euclidean spaces, it implies approximation and interpolation for maps from reduced Stein spaces; see [23, 26].

The *local extension approximation property* (LEAP), denoted by $\mathcal{L}_n$ for an n -dimensional target, asks for approximation while enlarging a planar source domain, with holomorphic dependence on auxiliary variables retained on every smaller parameter polydisc prescribed

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in advance. The given maps take values in a fixed neighborhood of each target point; the approximating maps may take values in the whole target. Definition 2.1 gives the precise formulation in terms of special planar disc pairs.

Theorem 1.2 (LEAP characterization). *A complex manifold is Oka if and only if it has the local extension approximation property.*

The main implication is $\text{LEAP} \Rightarrow \text{CAP}$: repeated planar extensions give approximation on polydiscs, and a real-face induction passes from polydiscs to arbitrary compact convex sets. Prescribing the retained parameter domain before approximation is essential because earlier source and spray variables become passive parameters at later stages; in LSAP, by contrast, the retained radius may depend on the spray [1, Definition 7.11 and Remark 7.12].

An n -dimensional complex manifold Y is *dominable* at $y \in Y$ if there is a holomorphic map $H : \mathbb{C}^n \rightarrow Y$ such that $H(0) = y$ and $dH_0 : \mathbb{C}^n \rightarrow T_y Y$ is an isomorphism. It is *strongly dominable* if it is dominable at every point. Every Oka manifold is strongly dominable [26, Proposition 7.1.9(a)]. Dominability at y in turn gives LEAP at y by lifting through a local inverse of a dominating entire map (Proposition 2.2). Strong dominability therefore gives LEAP everywhere, and Theorem 1.2 settles a central problem in Oka theory emphasized by Forstnerič and Lárusson [30, the paragraph following Theorem 1.11] and subsequently by Forstnerič [27, Section 3.3]:

Theorem 1.3. *A complex manifold is Oka if and only if it is strongly dominable.*

Alarcón and Forstnerič proved an Oka-1 principle under dominability outside a closed exceptional set of zero $(2n - 1)$ -dimensional Hausdorff measure [1, Theorem 2.2 and Corollary 2.5]. Their theorem gives approximation and jet interpolation for maps from open Riemann surfaces. In the particular case of dominability outside a proper closed complex analytic subset, their Corollary 2.5(b) gives Oka-1, whereas our Corollary 4.5 gives convex approximation in every finite source dimension under the same hypothesis. Further properties of Oka-1 manifolds and, in particular, of LSAP were developed by Forstnerič and Lárusson [30, Section 5].

For $d \geq 1$, let CAP_d denote convex approximation restricted to compact convex subsets of $\mathbb{C}^d$. Theorem 1.3 gives the following answer to [26, Problem 7.1.2(a)].

Corollary 1.4. *Let X be an n -dimensional complex manifold, where $n \geq 1$. Then X is Oka if and only if it satisfies CAP_n .*

A *finite Oka stratification* is a finite filtration

$$X = X_0 \supset X_1 \supset \cdots \supset X_k = \emptyset$$

by closed complex subvarieties such that each difference $X_i \setminus X_{i+1}$ is a complex manifold with Oka connected components [29, Definition 1]. Such a stratification implies strong dominability [29, Theorem 2], so Theorem 1.3 also gives the next result.

Corollary 1.5. *Every complex manifold with a finite Oka stratification is Oka.*

Kusakabe proved localization of the Oka property for Zariski-open covers and asked whether Euclidean-open covers suffice [41, Theorem 1.4 and Problem 1.6]. An entire map dominating an Oka neighborhood at a point also dominates the ambient manifold there. Hence Theorem 1.3 gives:

Corollary 1.6. *If every point of a complex manifold has an open Oka neighborhood, then the manifold is Oka.*

The proofs of Theorems 1.2 and 1.3, together with these three corollaries, are completed in Section 3.

LEAP also propagates across proper closed complex analytic subsets (Corollary 4.4), giving the following extension theorem for the Oka property.

Theorem 1.7. *Let X be a complex manifold and $A \subset X$ a proper closed complex analytic subset. If $X \setminus A$ is Oka, then X is Oka.*

The subset A need not be smooth or Oka. This theorem therefore answers the union problem [26, Problem 7.6.4] and strengthens Kusakabe’s Zariski localization theorem [41, Theorem 1.4]: one nonempty Zariski-open Oka subset suffices. In fact, Corollary 4.5 only requires X to be dominable at every point of $X \setminus A$ by entire maps with values in X ; their images may meet A .

The geometric applications use the following consequence (Corollary 4.7). Suppose $A \subset Y$ is a proper closed complex analytic subset and every $x \in Y \setminus A$ has a neighborhood $V \subset Y \setminus A$ with finitely many holomorphic maps

$$h_j : V \times \mathbb{C} \longrightarrow Y, \quad h_j(y, 0) = y,$$

whose vectors $\partial_t h_j(y, 0)$ span $T_y Y$ for every $y \in V$. Then Y is Oka. Each family is defined on the full product $V \times \mathbb{C}$, and its image may meet A . The case $A = \emptyset$ is included. These families first give LEAP outside A ; analytic propagation then gives LEAP everywhere, and Theorem 1.2 applies.

The analytic proofs extend constructions developed for surfaces. Xie and Zhao constructed Oka K3 surfaces and developed a period-theoretic framework for studying their Oka loci [52]; Wang, Xie, and Zhao treated genus-one K3 surfaces and generalized Kummer surfaces [50]. The proof that every complex K3 surface is Oka [46] builds on this framework and directly inspired the present work. Our LEAP-to-CAP argument extends the polynomial source coordinates, polydisc extension, real-face induction, and interface gluing of that proof to arbitrary target dimension. The source-side constructions are unchanged; the chart sprays have n fiber variables, retained as passive parameters when the source is enlarged. Lemma 3.1 gives a different proof of the real-parameter complexification step. For surfaces, the dominability implication is already a special case of [46, Proposition 2.2].

Propagation across analytic subsets requires n independent directions, extending the two-direction argument of [46, Corollary 2.10]. Section 4 enlarges coordinate discs and recenters the resulting families to obtain the exact normalization $h_{j,\nu}(y, 0) = y$. We adapt the scalar-substitution method of [4, Lemma 4.3 and Section 4.4]; since the scalar domains have radii tending to infinity, their substitutions are approximated by polynomials before the family index is chosen.

Geometric applications. A smooth complex projective manifold X is *rationally connected* if a general pair of points of X lies on the image of a holomorphic map $\mathbb{P}^1 \rightarrow X$. It is *Fano* if K_X^{-1} is ample. Smooth complex Fano manifolds are rationally connected [5, 40]. Deformations of very free rational curves give our first application.

Theorem 1.8. *Every smooth complex projective rationally connected manifold is Oka. In particular, every smooth complex Fano manifold is Oka.*

Consequently, every smooth hypersurface of degree $d \leq n$ in $\mathbb{P}^n$ is Oka. This gives a uniform affirmative answer throughout the Fano range to the low-degree hypersurface problem posed by Forstnerič and Lárusson [28, Problem E, pp. 33–34].

Campana and Winkelmann constructed dense entire curves in rationally connected projective manifolds with prescribed countable subsets in their images [6]. The Oka-1 conjecture for these manifolds [1, Conjecture 9.1] was proved in [16, Corollary 1.8]; related analytic constructions of rational curves appear in [17]. Earlier Oka results include smooth quartic hypersurfaces of dimension at least three [37] and smooth Fano threefolds [4]. Section 5 constructs holomorphic families of rational curves with independent tangent directions and applies Corollary 4.7 to obtain the full Oka property.

For complements, known results include Kusakabe’s theorem on complements of compact polynomially convex sets in $\mathbb{C}^n$, $n > 1$ [42, Theorem 1.2 and Corollary 1.3], classes of

unbounded convex sets [33], and periodic-transport constructions [15]. In the projective setting, Section 6 proves a criterion using a rational curve $f : \mathbb{P}^1 \rightarrow X$ with $f^{-1}(D) \subseteq \{\infty\}$ and ample $f^*T_X(-\log D)$, where D is a reduced simple normal crossings divisor. Fixed-contact deformations give entire-curve families outside a proper algebraic subset, and the analytic removability theorem then gives the Oka property of $X \setminus D$. Chen–Zhu’s existence results [8] yield applications to ample rationally connected divisors, complete intersections, and several boundary components under the hypotheses stated there.

A *holomorphic spray* on Y is a holomorphic map $s : E \rightarrow Y$ from a holomorphic vector bundle $E \rightarrow Y$ such that $s(0_y) = y$. It is *dominating* if $d(s|_{E_y})_{0_y} : E_y \rightarrow T_y Y$ is surjective for every y . A manifold admitting such a spray is *holomorphically elliptic*; Gromov’s theorem implies that it is Oka [36]. For smooth hypersurface complements we obtain an exact degree criterion.

Theorem 1.9. *Let $D \subset \mathbb{P}^n$ be a smooth hypersurface of degree $d \geq 1$, where $n \geq 1$. The following conditions are equivalent:*

- (i) $\mathbb{P}^n \setminus D$ is Oka;
- (ii) $\mathbb{P}^n \setminus D$ is holomorphically elliptic;
- (iii) $d \leq n + 1$.

The cubic cases in the Oka range were first settled by Xie [51, Theorem 1.1]; in particular, the plane cubic case answered a question of Forstnerič and Lárusson [28, Problem B, p. 33]. More broadly, they asked whether the complement of a generic low-degree hypersurface in projective space is Oka [28, Problem E, pp. 33–34]. Theorem 1.9 settles this problem for every smooth hypersurface and gives the sharp threshold $d \leq n + 1$.

At the anticanonical degree $d = n + 1$, the determinant of $T_{\mathbb{P}^n}(-\log D)$ is trivial, so its pullback to a rational curve cannot be ample. Section 7 instead uses successive polar equations to construct local holomorphic families of lines meeting D in at most two distinct points, and then parametrizes their complements by entire curves. For $d \geq n + 2$, the rank obstruction of Carlson–Griffiths [7, Corollary (6.11)] excludes the Oka property. In particular, the theorem includes the complements of smooth quartic K3 surfaces in $\mathbb{P}^3$ and smooth quintic threefolds in $\mathbb{P}^4$.

Complete vertical flows give the abelian-fibration theorem: a smooth complex projective manifold fibered over $\mathbb{P}^1$ with a section and abelian fibers over a nonempty Zariski open set is Oka. The construction also gives the entire map in Theorem 8.1 and does not require the canonical bundle to be trivial. Section 8 applies the theorem to smooth Schoen fiber products, which are Oka but not rationally connected. More generally, the projective manifolds covered by Theorems 1.8 and 8.1 are holomorphically elliptic by [31, Theorem 1.1]; see Corollary 8.3.

Separately, Alpöge constructed the first known integrable complex structures on the smooth six-sphere [3]. Du–Guo–Kusakabe–Xie [13] and, independently, Chen [9] proved that every member of Alpöge’s family is Oka.

Finally, applying Theorem 1.7 to a normalized evaluation fiber and then using the Oka fiber bundle theorem gives:

Proposition 1.10. *For every $d \geq 1$, the manifold $\text{Hol}_d(\mathbb{P}^1, \mathbb{P}^1)$ is Oka.*

The proof appears in Section 9; the degree-three case is related to the results of [38, 51]. Further consequences of the analytic criteria are discussed in [14].

Organization. Sections 2–3 prove the LEAP characterization by passing from planar extension to CAP, and derive the strong dominability criterion. Section 4 proves propagation across analytic subsets and the entire-curve-family criterion. Sections 5–7 treat rationally connected manifolds and divisor complements; Section 8 develops the abelian-fibration theorem and its Schoen Calabi–Yau application. Section 9 concludes with spaces of rational maps. Appendices A

and [B](#) supply the planar enlargement and polydisc approximation arguments used in the main proofs.

2. THE LOCAL EXTENSION APPROXIMATION PROPERTY

2.1. Definition and elementary consequences. For a compact set K , the notation $\text{Op } K$ denotes an unspecified open neighborhood of K , and $\mathcal{O}(\text{Op } K, Y)$ denotes maps to Y holomorphic on such a neighborhood. Let d_Y be a complete distance inducing the topology of the target manifold Y ; its existence follows from the complete Riemannian metric theorem of Nomizu–Ozeki [44, Theorem 1]. Completeness is used only in the exhaustion at the end of Appendix [B](#).

For $b > 0$, we write

$$\Delta_b = \{\zeta \in \mathbb{C} : |\zeta| < b\}, \quad \Delta = \Delta_1, \quad \Delta_b^n = \{v \in \mathbb{C}^n : \|v\|_\infty < b\}.$$

Here $|\cdot|$ denotes the Euclidean norm and $\|\cdot\|_\infty$ the maximum norm. Throughout the analytic arguments, n denotes the target dimension, m the total complex source dimension, and q the number of passive variables when these are displayed separately. In CAP_n , by contrast, the subscript specifies the source dimension at which convex approximation is assumed. For a subset E of a complex Euclidean space, $\text{int } E$ denotes its interior in that space; $E \Subset U$ means that $\overline{E}$ is compact and contained in U . The closed polydisc $\overline{\Delta}_b^m$ consists of the points $z \in \mathbb{C}^m$ with $|z_j| \leq b$ for $1 \leq j \leq m$.

In planar approximation, *passive parameters* are the variables other than the planar variable being extended: holomorphic dependence on these variables is preserved on a prescribed smaller polydisc. The dimension of each such parameter polydisc is specified at its introduction.

Definition 2.1 (Local extension approximation property (LEAP)). A *special planar disc pair* $D \subset D'$ consists of closed topological discs $D, D' \subset \mathbb{C}$ with piecewise smooth boundaries such that $D' = D \cup E$ for another closed disc E , where $D \cap E$ is a proper boundary arc of both D and E . An n -dimensional complex manifold Y has the *local extension approximation property* (LEAP), denoted by $\mathcal{L}_n$, at y if there is an open neighborhood $V \ni y$ such that, for every special disc pair $D \subset D'$, every integer $q \geq 0$, every pair of closed polydiscs

$$P_0 \Subset \text{int } P \subset \mathbb{C}^q$$

and $f \in \mathcal{O}(\text{Op}(D \times P), V)$, the map f can be approximated uniformly on $D \times P_0$ by maps in $\mathcal{O}(\text{Op}(D' \times P_0), Y)$. Only the given map is required to take values in V ; the approximating maps take values in Y and may leave V . The set V is a *witness neighborhood* for LEAP at y . A witness neighborhood contained in a target coordinate chart is called a *coordinate witness neighborhood*. We say that Y has LEAP if it has $\mathcal{L}_n$ at every point.

By shrinking a witness neighborhood about its distinguished point, we may always assume that it is a coordinate witness neighborhood.

The Euclidean model. The simplest instance of Definition 2.1 is $Y = \mathbb{C}^n$. Indeed, if

$$f \in \mathcal{O}(\text{Op}(D \times P), \mathbb{C}^n),$$

then $D \times P$ is polynomially convex. The Cartan–Oka–Weil theorem, applied componentwise, therefore approximates f uniformly on $D \times P$ by entire maps $\mathbb{C}^{1+q} \rightarrow \mathbb{C}^n$ [26, Theorem 2.8.4]. These maps are, in particular, holomorphic near $D' \times P_0$. Thus $\mathbb{C}^n$ has property $\mathcal{L}_n$ at every point.

Proposition 2.2. *If Y is dominable at y , then Y has LEAP at y .*

Proof. If $H : \mathbb{C}^n \rightarrow Y$ dominates at y , the inverse function theorem gives a biholomorphic restriction $H|_W : W \rightarrow V$. Lift an admissible map into W , apply the Euclidean model, and compose the approximants with the globally defined map H . $\square$

LEAP is a parameter-uniform strengthening of the local spray approximation property (LSAP) [1, Definition 7.11]. Although the domination condition in LSAP may be omitted [1, Remark 7.12(B)], LSAP retains a ball whose radius may depend on the spray, whereas $\mathcal{L}_n$ retains every prescribed $P_0 \in \text{int } P$. With the planar pair and the initial parameter domain fixed, the quantifier order is schematically

$$\text{LEAP} : \quad \forall P_0 \in \text{int } P \quad \forall f \quad \forall \epsilon > 0 \quad \exists g,$$

$$\text{LSAP} : \quad \forall F \quad \exists r \in (0, 1) \quad \forall \epsilon > 0 \quad \exists G.$$

Here f and F range over the admissible maps and sprays, and approximation is required on $D \times P_0$ and $D \times r\overline{B^N}$, respectively, with B^N the unit ball in $\mathbb{C}^N$. In the iteration below, source and spray variables already in use become passive parameters for the next extension, so the retained core must be prescribed before approximation.

The auxiliary parameters in LEAP are complex and enter holomorphically; the usual parametric Oka principle with topological parameters follows once CAP is established [24, Theorem 1.1]. For $n = 2$, Definition 2.1 is precisely property $\mathcal{L}$ of [46, Definition 2.1].

Unlike Kusakabe's Ell_1 criterion, whose sprays are defined on the entire parameter fiber, LEAP begins with a local planar extension operation on target neighborhoods [41, Theorems 1.3 and 2.2].

The analytic core of the characterization is the following amplification theorem.

Theorem 2.3. *Let Y be an n -dimensional complex manifold. If Y has LEAP, then Y has CAP.*

The remaining subsections establish polydisc approximation, and the passage to CAP is completed at the end of Section 3. For $n = 2$, the result is [46, Theorem 3.1]. The polynomial source coordinates and polydisc extension are taken from [46, Lemma 3.3 and Proposition 3.4], while the real-interface splitting is taken from [46, Lemmas 4.4–4.5]. For arbitrary n , these source-side constructions are unchanged; the dimensional modification is that the chart sprays carry n fiber variables. Theorem 2.3 is therefore an extension in scope of the surface theorem.

2.2. Chart sprays and planar splitting. Our gluing construction needs holomorphic fiber coordinates along the graph of a map. They are supplied by the following standard local thickening lemma.

Lemma 2.4. *Let $f : S \rightarrow Y$ be holomorphic, where S is Stein, f^*TY is holomorphically trivial on S , and $K \Subset S$. There are an open neighborhood $U \subset S$ of K , a number $\beta > 0$, and a holomorphic map*

$$f^\sharp : U \times \Delta_\beta^n \longrightarrow Y, \quad f^\sharp(z, 0) = f(z),$$

such that

$$(z, v) \longmapsto (z, f^\sharp(z, v))$$

is biholomorphic onto its image. For every $K' \Subset U$ and $0 < \beta' < \beta$, there is an open neighborhood of the image of $K' \times \overline{\Delta}_{\beta'}^n$ on which the inverse is holomorphic.

A precursor of this local-spray construction appears in [23, Lemma 3.2]; the graph-chart form is [46, Lemma 2.6]. In the applications below, the source is either a contractible Stein domain or a product of a planar domain with a polydisc. A planar domain has the homotopy type of a one-dimensional CW complex, and taking its product with a polydisc does not change that homotopy type. Complex vector bundles on these sources are therefore topologically trivial. The Oka–Grauert principle gives the required holomorphic triviality, as in the proof of the cited lemma; the graph-chart construction then applies with n fiber variables.

In these fiber coordinates, the linear step in Cartan-type gluing is an additive Cousin splitting. A related bounded additive splitting in the Cartan-pair setting appears in [21, Lemma 4.6]; see also [26, Lemma 5.8.2]. The precise planar statement used here, including

passive holomorphic parameters and an operator bound independent of their number and range, is [46, Lemma 2.4]. We record the quantitative consequence needed below.

For an open set $U \subset \mathbb{C}^m$, let

$$H^\infty(U, \mathbb{C}^N) = \{f \in \mathcal{O}(U, \mathbb{C}^N) : \sup_{z \in U} \|f(z)\|_\infty < \infty\}$$

be the Banach space of bounded holomorphic maps, equipped with the supremum norm $\|f\|_{\infty, U} = \sup_{z \in U} \|f(z)\|_\infty$; matrix norms are the induced operator norms. In Lemmas 2.5 and 2.6, the passive parameter polydisc P is open. The closed parameter polydiscs used in applications are compactly contained in such a P .

Lemma 2.5. *Suppose that U_A, U_B are bounded planar open sets and that the closed sets $\overline{U_A \setminus U_B}$ and $\overline{U_B \setminus U_A}$ are disjoint. For their overlap $C = U_A \cap U_B$, every open passive parameter polydisc P , and every $N \geq 1$, there are bounded linear operators*

$$R_A : H^\infty(C \times P, \mathbb{C}^N) \longrightarrow H^\infty(U_A \times P, \mathbb{C}^N), \quad R_B : H^\infty(C \times P, \mathbb{C}^N) \longrightarrow H^\infty(U_B \times P, \mathbb{C}^N)$$

such that

$$(2.1) \quad R_A u - R_B u = u, \quad \|R_A u\|_{\infty, U_A \times P} + \|R_B u\|_{\infty, U_B \times P} \leq C_0 \|u\|_{\infty, C \times P}.$$

They preserve holomorphic dependence on all passive parameters. The constant C_0 depends only on the planar pair and the chosen cutoff; in particular, it is independent of N and of the radii and dimension of the passive parameter polydisc P .

Proof. This is [46, Lemma 2.4]. Its Cauchy-integral construction acts componentwise. If χ is the separating cutoff and $d \geq \text{diam}(U_A \cup U_B)$, it gives

$$C_0 = 2 + 4d \|\partial_{\bar{s}} \chi\|_\infty,$$

independently of the range dimension and of the radii and dimension of P . $\square$

The next lemma turns a small transition error into exact agreement of two sprays. It records the quantitative contraction estimate used in [46, Lemma 2.8]; we include the short proof because its explicit bounds are used below.

Lemma 2.6. *Let U_A, U_B, C, R_A, R_B , and C_0 be as in Lemma 2.5, let P be an open polydisc, and let*

$$F_A \in \mathcal{O}(U_A \times P \times \Delta_\rho^n, Y), \quad F_B \in \mathcal{O}(U_B \times P \times \Delta_\rho^n, Y).$$

Assume that F_A is fiberwise injective and that the transition

$$(2.2) \quad \gamma(s, t, v) = F_A(s, t, \cdot)^{-1}(F_B(s, t, v)) = v + E(s, t, v)$$

is defined and holomorphic on $C \times P \times \Delta_\rho^n$. Set

$$e = \|E\|_\infty, \quad \kappa = \|D_v E\|_\infty,$$

where both norms are taken on this product. If

$$(2.3) \quad 2C_0 e < \rho, \quad C_0 \kappa < \frac{1}{2}$$

then there are

$$a \in H^\infty(U_A \times P, \mathbb{C}^n), \quad b \in H^\infty(U_B \times P, \mathbb{C}^n)$$

such that

$$(2.4) \quad F_A(s, t, a(s, t)) = F_B(s, t, b(s, t)), \quad \|a\|_{\infty, U_A \times P} + \|b\|_{\infty, U_B \times P} \leq C_0 e < \rho/2.$$

Consequently

$$F(s, t) = \begin{cases} F_A(s, t, a(s, t)), & s \in U_A, \\ F_B(s, t, b(s, t)), & s \in U_B \end{cases}$$

is holomorphic on $(U_A \cup U_B) \times P$. If $L_A \subset U_A \times P$ is compact, then

$$\sup_{L_A} d_Y(F, F_A(\cdot, 0)) \leq \omega_{A, L_A}(C_0 e),$$

where ω_{A,L_A} is a continuity modulus of F_A in its fiber variable.

Proof. On the Banach space $X = H^\infty(C \times P, \mathbb{C}^n)$, consider

$$(2.5) \quad \mathcal{T}(u)(s, t) = E(s, t, R_B u(s, t)).$$

On the ball $\|u\|_{\infty, C \times P} \leq 2e$ in X , the first inequality in (2.3) ensures that $R_B u$ remains in Δ_ρ^n , while

$$(2.6) \quad \|\mathcal{T}(u)\|_{\infty, C \times P} \leq e, \quad \|\mathcal{T}(u) - \mathcal{T}(\tilde{u})\|_{\infty, C \times P} \leq C_0 \kappa \|u - \tilde{u}\|_{\infty, C \times P}.$$

Hence $\mathcal{T}$ has a fixed point u with $\|u\|_{\infty, C \times P} \leq e$. Set $a = R_A u$ and $b = R_B u$. Then

$$a - b = u = E(s, t, b), \quad a = \gamma(s, t, b),$$

and Lemma 2.5 gives

$$\|a\|_{\infty, U_A \times P} + \|b\|_{\infty, U_B \times P} \leq C_0 e.$$

Thus $F_A(s, t, a) = F_B(s, t, b)$ on the overlap, proving all the assertions. $\square$

2.3. Planar enlargement. The next lemma uses the radial-spoke and sector-filling construction of [46, Proposition 2.9], with the retained parameter polydisc prescribed in advance. It combines Mergelyan approximation with the spray framework of Gromov [36, Section 1.1.B] and the local-spray and splitting techniques used in [23, Lemma 2.1, Proposition 3.1, and Lemma 3.2]. Appendix A gives the construction with n spray variables.

The disc case will be used to construct expanding families in Section 4; the annular case will extend the maps expressed in the source coordinates of Subsection 2.4.

Lemma 2.7. *Let $r > 0$,*

$$K = \overline{\Delta}_r \quad \text{or} \quad K = \{s \in \mathbb{C} : r_- \leq |s| \leq r\}, \quad 0 < r_- < r,$$

and suppose that f is holomorphic near $K \times P$ for a closed polydisc $P \subset \mathbb{C}^q$, where $q \geq 0$. Suppose in addition that for every $\zeta \in \partial \Delta_r$ there is a coordinate witness neighborhood V_ζ such that

$$f(\{\zeta\} \times P) \Subset V_\zeta.$$

For every $R > r$, every closed polydisc $P_0 \Subset \text{int } P$, and $\epsilon > 0$, there is

$$F \in \mathcal{O}(\text{Op}(K_R \times P_0), Y), \quad \sup_{K \times P_0} d_Y(F, f) < \epsilon,$$

where $K_R = \overline{\Delta}_R$ in the disc case and $K_R = \{r_- \leq |s| \leq R\}$ in the annular case.

The proof, including the verification that the construction is unchanged with n spray variables, is given in Appendix A.

2.4. Polynomial source coordinates and polydisc approximation. The passage to polydisc approximation follows [46, Proposition 3.4]. The polynomial coordinates in Lemma 2.8 have transverse fibers of small diameter on the joining annulus and give a change of coordinates close to the identity near the initial polydisc. Planar enlargement, gluing, and exhaustion then yield Proposition 2.9.

Lemma 2.8. *Fix $m \geq 2$ and*

$$1 < r_m < \cdots < r_2 < r_1, \quad 0 < a < 1, \quad r_j < a r_{j-1} \quad (2 \leq j \leq m).$$

Set

$$P_j(w) = (w/r_j)^d, \quad H_d(x) = (a x_m, P_2(x_2) - a x_1, \dots, P_m(x_m) - a x_{m-1}).$$

This is a polynomial automorphism. For $y = H_d(x)$, its inverse is given recursively by

$$x_m = y_1/a, \quad x_{j-1} = (P_j(x_j) - y_j)/a \quad (j = m, m-1, \dots, 2).$$

There exist $C \geq 1$, $q \in (0, 1)$, and an integer $d_0 \geq 2$, depending only on the fixed data, such that for every pair of integers $d \geq d_0$ and $J \geq 1$, the set $E_{d,J} = H_d^{-J}(\bar{\Delta}_1^m)$ satisfies

$$(2.7) \quad \max_{j \geq 2} |P_j(x_j)| \leq C \max(1, |x_1|).$$

If also $|x_1| > r_1$, then all x_j are nonzero and

$$(2.8) \quad \left| \frac{H_d(x)_j}{ax_{j-1}} \right| \leq q \quad (2 \leq j \leq m).$$

By (2.8), the numbers

$$\xi_j = \log(P_j(x_j)/(ax_{j-1})) \quad (2 \leq j \leq m)$$

are defined using the same logarithm branch centered at 1, and satisfy $|\xi_j| \leq B = -\log(1 - q)$.

With

$$D = d^{m-1}, \quad \rho_1 = 1, \quad \rho_j = r_j(a\rho_{j-1})^{1/d},$$

there is a single-valued function s , holomorphic on an open neighborhood of $E_{d,J} \cap \{|x_1| > r_1\}$, such that

$$(2.9) \quad x_j = \rho_j s^{d^{m-j}} \exp\left(\sum_{k=2}^j \frac{\xi_k}{d^{j-k+1}}\right), \quad x_1 = s^D.$$

The proof is in [46, Lemma 3.3]; for later use we recall the inverse formula and the two estimates from it.

The single-valued coordinate in (2.9) is

$$s = \frac{x_m}{\rho_m} \exp\left(-\sum_{k=2}^m \frac{\xi_k}{d^{m-k+1}}\right).$$

The next estimates quantify the variation in the transverse parameters on the joining annulus. Write $\Phi_d(s, \xi)$ for the right side of (2.9), now allowing ξ to vary independently. Given $\ell > 0$, fix

$$r_1 < c_0 < c_1 < c_2 < c_3 < r_1 + \ell$$

and $\Xi_2 = \{\xi : |\xi_j| \leq B + 3\}$. There are constants C_1, C_2 depending only on the fixed data r_1, a, B, c_3 , and independent of d, J, s, ξ , such that, for all sufficiently large d , on the annulus $c_0 \leq |s|^D \leq c_3$ and for $\xi \in \Xi_2$,

$$(2.10) \quad |\Phi_{d,j}(s, \xi)| \leq r_j e^{C_1/d} \quad (j \geq 2),$$

$$(2.11) \quad \sup_{\xi, \tilde{\xi} \in \Xi_2} \left\| \Phi_d(s, \xi) - \Phi_d(s, \tilde{\xi}) \right\|_{\infty} \leq C_2/d.$$

Equations (2.10) and (2.11) are the estimates in [46, equation (3.8)], with the transverse polydisc fixed at Ξ_2 .

The change of source coordinates is also close to the identity near the original polydisc. With

$$L(x) = (ax_m, -ax_1, \dots, -ax_{m-1}),$$

first fix J with $a^J R_+ < 1$ for a prescribed $R_+ > 1$, and set

$$(2.12) \quad h_d = H_d^{-J} L^J.$$

On a fixed neighborhood of the unit polydisc, $h_d \rightarrow \text{id}$. More precisely, choose $1 < b_0 < b_1 < r_m$ and $\theta = b_1/r_m < 1$. The estimate proving [46, equation (3.7)] gives, for large d ,

$$(2.13) \quad \|h_d - \text{id}\|_{\bar{\Delta}_{b_0}^m} \leq \theta^d \sum_{i=1}^J a^{-i} \longrightarrow 0.$$

The constant J is fixed before d tends to infinity.

Combining these coordinates with Lemma 2.7 gives the polydisc approximation needed to start the convex-set induction in Section 3.

Proposition 2.9. *If Y has LEAP at every point, then for every $m \geq 1$, every closed polydisc $Q \subset \mathbb{C}^m$, and $f \in \mathcal{O}(\text{Op } Q, Y)$, the map f can be approximated uniformly on Q by maps in $\mathcal{O}(\mathbb{C}^m, Y)$.*

Appendix B proves this: for $m \geq 2$, polynomial source coordinates reduce the problem to planar enlargement with $O(1/d)$ -thin transverse fibers, compositional splitting glues the resulting sprays, and a summable-error exhaustion yields an entire map.

3. FROM LEAP TO CONVEX APPROXIMATION

We follow the face-removal induction of [46, Section 4, proof of Theorem 3.1]: the half-space constraints of an enclosing polytope are removed one at a time inside a larger polydisc. Lemmas 3.2–3.4 use [46, Lemmas 4.2–4.5] with n spray variables. For Lemma 3.1, corresponding to [46, Lemma 4.1], we give a different Stein-neighborhood proof.

For each integer $r \geq 0$, let A_r be the following approximation property: for every compact convex set $K \subset \mathbb{C}^r$ and every closed polydisc Q , each map to Y holomorphic near $K \times Q$ can be approximated uniformly there by maps entire in all source variables. Proposition 2.9 gives A_0 . Approximation on real faces and gluing across a single hyperplane will give $A_{r-1} \Rightarrow A_r$.

3.1. Real parameters and approximation on real faces. On a manifold target, finite derivative accuracy is measured in fixed target charts around the compact image of a reference map. Take finitely many compact source pieces K_ν and target charts ψ_ν whose domains contain $H(K_\nu)$ with a positive margin. For a map G with $\psi_\nu \circ G$ defined on K_ν , write

$$\|G - H\|_{C^j(K)} = \max_{\nu} \max_{|\mu| \leq j} \sup_{K_\nu} |D^\mu(\psi_\nu \circ G - \psi_\nu \circ H)|.$$

Derivatives are taken in the displayed real and complex source variables; on a real face they are tangential. Different finite families of charts give equivalent local notions of convergence. For maps jointly holomorphic on a fixed complex neighborhood, uniform approximation on a slightly larger compact neighborhood gives approximation in any fixed finite number of derivatives on the smaller set by Cauchy estimates.

To turn the real-parameter families used in Lemma 3.3 into holomorphic families, we use the following arbitrary-target analogue of [46, Lemma 4.1]. The proof below proceeds through a Stein neighborhood of the parameterized graph and Banach-valued polynomial approximation, followed by holomorphic retraction.

Lemma 3.1. *Let $I \subset \mathbb{R}$ be a compact interval and $C \subset \mathbb{C}^k$ a compact convex set, with open neighborhoods $J \subset \mathbb{R}$ and $U \subset \mathbb{C}^k$, respectively. Suppose that*

$$H : J \times U \longrightarrow Y$$

is jointly smooth and that $H_s = H(s, \cdot)$ is holomorphic for every real $s \in J$. For every integer $j \geq 0$ and every $\epsilon > 0$, there exist complex open neighborhoods $V \supset I$, $U_0 \supset C$ and

$$F \in \mathcal{O}(V \times U_0, Y), \quad \|F - H\|_{C^j(I \times C)} < \epsilon.$$

More generally, the approximation may be required on any prescribed compact product $I_1 \times C_1$ satisfying

$$I \Subset \text{int } I_1 \Subset J, \quad C \Subset \text{int } C_1 \Subset U,$$

where I_1 is an interval and C_1 is convex. The thickness of the complex neighborhood may depend on F .

Proof. Choose a compact interval $I^+ = [a, b] \Subset J$, a compact convex set C^+ , and a bounded convex open set X such that

$$(3.1) \quad I \Subset \text{int } I^+, \quad C \Subset \text{int } C^+ \Subset X \Subset U.$$

Apply [32, Corollary 2.2] with parameter set $I^+ \subset \mathbb{R}$, Stein base X , submersion $X \times Y \rightarrow X$, the compact $\mathcal{O}(X)$ -convex set C^+ , and the holomorphic sections $q \mapsto (q, H_s(q))$. This gives a Stein neighborhood Θ of the graph

$$(3.2) \quad \Sigma = \{(s, q, H_s(q)) : s \in I^+, q \in C^+\} \subset \mathbb{C} \times X \times Y.$$

The Remmert–Bishop–Narasimhan embedding theorem and the Docquier–Grauert retraction theorem, in the form used in [19, proof of Lemma 3], give

$$(3.3) \quad e : \Theta \hookrightarrow \mathbb{C}^N, \quad e(\Theta) \text{ closed}, \quad \rho : \Omega \longrightarrow e(\Theta), \quad \rho|_{e(\Theta)} = \text{id},$$

with $\Omega \subset \mathbb{C}^N$ open and ρ holomorphic.

Choose an open set U_1 with $C \Subset U_1 \Subset \text{int } C^+$. Compactness of (3.2) gives

$$(3.4) \quad g(s, q) = e(s, q, H_s(q)), \quad g \in C^\infty(I^+, H^\infty(U_1, \mathbb{C}^N)).$$

Write $\mathcal{B} = H^\infty(U_1, \mathbb{C}^N)$ and $L = j + 1$, so that the derivatives $\partial_s^\ell g$ with $0 \leq \ell < L$ are exactly the s -derivatives of g up to order j . For $x = (s - a)/(b - a)$, form the $\mathcal{B}$ -valued polynomial

$$(3.5) \quad Q_\nu(s) = \sum_{k=0}^{\nu} \partial_s^L g \left(a + \frac{k(b-a)}{\nu} \right) \binom{\nu}{k} x^k (1-x)^{\nu-k}.$$

The binomial weights in (3.5) have mean x and variance $x(1-x)/\nu$. Split the sum at $|a + k(b-a)/\nu - s| = (b-a)\nu^{-1/4}$. Uniform continuity of $\partial_s^L g$ controls the terms within this distance, while the variance estimate makes the remaining total weight tend to zero. Thus, in the Banach norm,

$$(3.6) \quad \varepsilon_\nu := \sup_{I^+} \|Q_\nu - \partial_s^L g\|_{\mathcal{B}} \longrightarrow 0.$$

Integrating (3.5) L times and choosing the integration constants so that the first $L - 1$ derivatives at a agree with those of g , set the following polynomial in s , whose coefficients are holomorphic in q :

$$(3.7) \quad P_\nu(s) = \sum_{\ell=0}^{L-1} \frac{(s-a)^\ell}{\ell!} \partial_s^\ell g(a) + \int_a^s \frac{(s-t)^{L-1}}{(L-1)!} Q_\nu(t) dt.$$

By (3.7), for $0 \leq \ell < L$ the difference $\partial_s^\ell (P_\nu - g)$ is the $(L - \ell)$ -fold primitive of $Q_\nu - \partial_s^L g$ that vanishes to order $L - \ell$ at $s = a$; hence (3.6) implies

$$(3.8) \quad \sup_{I^+} \|\partial_s^\ell (P_\nu - g)\|_{\mathcal{B}} \leq \frac{(b-a)^{L-\ell}}{(L-\ell)!} \varepsilon_\nu.$$

Equation (3.8) and Cauchy estimates across the fixed margin $C \Subset U_1$ yield

$$(3.9) \quad \max_{\ell+|\beta| \leq j} \sup_{I \times C} |\partial_s^\ell \partial_q^\beta (P_\nu - g)| \longrightarrow 0.$$

For large ν , the case $\ell = 0$ of (3.8) places $P_\nu(I \times \overline{U_0})$ in Ω for some open U_0 with $C \Subset U_0 \Subset U_1$. By compactness of I , there is then a complex neighborhood $V_\nu \supset I$ with $P_\nu(V_\nu \times U_0) \subset \Omega$. Consequently

$$F_\nu = \pi_Y \circ e^{-1} \circ \rho \circ P_\nu \in \mathcal{O}(V_\nu \times U_0, Y).$$

On fixed charts around the compact image, (3.3), (3.4), (3.9), and the chain rule give

$$\|F_\nu - H\|_{C^j(I \times C)} \leq C_j \max_{\ell+|\beta| \leq j} \sup_{I \times C} |\partial_s^\ell \partial_q^\beta (P_\nu - g)| \longrightarrow 0.$$

The margins in (3.1) can be taken around any prescribed compact product inside $J \times U$. $\square$

After complexifying the real parameter, we use [46, Lemma 4.2] to make the approximation entire in that variable as well. A Riemann map places the parameter in a polydisc factor; it is approximated by a polynomial after the entire target-valued approximant has been fixed.

Lemma 3.2. *Suppose that Y satisfies A_r for an integer $r \geq 0$. Let $I \subset \mathbb{R}$ be a compact interval, let $C \subset \mathbb{C}^r$ be compact and convex, and let $P \subset \mathbb{C}^q$ be a closed polydisc, where $q \geq 0$. For every $j \geq 0$ and $\epsilon > 0$, each map H holomorphic near $I \times C \times P$ admits an entire map $\tilde{H}$ in all displayed complex variables such that*

$$\|\tilde{H} - H\|_{C^j(I \times C \times P)} < \epsilon.$$

Proof. Write the map as $H(s, y, u)$. Choose a bounded simply connected neighborhood V of I on which it is holomorphic, a Riemann map $\varphi : V \rightarrow \Delta$, and $0 < \rho < 1$ with $\varphi(I) \Subset \Delta_\rho$. Apply A_r to

$$\hat{H}(\zeta, y, u) = H(\varphi^{-1}(\zeta), y, u),$$

treating $\overline{\Delta}_\rho \times P$ as a passive polydisc. After retaining slightly larger compact domains, Cauchy estimates give an entire map $E(\zeta, y, u)$ for which $E(\varphi(s), y, u)$ approximates H with the prescribed finite derivative accuracy on $I \times C \times P$. Fix E . By Runge's theorem [19, Theorem 2], choose a polynomial p approximating φ , together with the required derivatives, on I . Then $E(p(s), y, u)$ is entire in all variables, and the chain rule shows that it approximates H as closely as desired. $\square$

The convex slices of a real face vary with its real coordinate. We use the construction of [46, Lemma 4.3], which joins their separate approximants into one family on a common domain. Preserving the endpoint maps exactly makes these local paths fit together.

Lemma 3.3. *Assume that A_{r-1} holds, where $r \geq 1$. Let $F \subset \mathbb{R} \times \mathbb{C}^{r-1}$ be compact and convex and P a closed polydisc. Every holomorphic map defined near $F \times P$ can be approximated on this set by entire maps on $\mathbb{C}^r \times \mathbb{C}^{\dim P}$, with control of tangential derivatives of any fixed finite order.*

Proof. Let f be the given map and write the variables as (t, y, u) , with t real on the face. The proof has four stages: approximate finitely many slices, place the joining paths on a common domain, repair their endpoints exactly, and then assemble and complexify the resulting family. Rescaling provides the common domain, while endpoint repair makes the local paths concatenate exactly.

Slice approximations. Thicken F and P slightly within the original domain of definition. Denote the projection of F onto the t -axis by I and the slices by F_t . For each $t_0 \in I$, choose a compact convex neighborhood C_{t_0} of the whole slice F_{t_0} and a complex disc D_{t_0} about t_0 such that the original map is defined near $D_{t_0} \times C_{t_0} \times P$. The inclusion $F_{t_0} \subset \text{int } C_{t_0}$ persists under perturbation: if not, there would be

$$t_\nu \rightarrow t_0, \quad (t_\nu, y_\nu) \in F, \quad y_\nu \notin \text{int } C_{t_0}.$$

Compactness gives a subsequence with $(t_\nu, y_\nu) \rightarrow (t_0, y) \in F$, whereas the closedness of $\mathbb{C}^{r-1} \setminus \text{int } C_{t_0}$ gives

$$y \in F_{t_0} \setminus \text{int } C_{t_0} = \emptyset.$$

A finite subdivision of I therefore gives products $D_i \times C_i \times P$ with reserved margins, such that on each subinterval the whole slice F_t lies uniformly inside its assigned C_i . Near each subdivision point, choose

$$F_t \Subset \text{int } C^-, \quad C^- \Subset \text{int } C, \quad C \Subset \text{int}(C_i \cap C_{i+1}),$$

where C^-, C are compact convex intermediate sets with nonempty interior. At a subdivision point, choose a small complex disc D_b about that point so that $\overline{D}_b \times C \times P$ lies, with positive

margins, in both adjacent domains of holomorphy. Also choose a common large polydisc B containing all the C_i .

Apply A_{r-1} , treating the t -disc and the u -polydisc as passive parameters, on fixed compact complex products containing the assigned portion of $F \times P$ and each adjacent product $\overline{D}_b \times C \times P$ in their interiors. The reserved margins allow these compact products to be enlarged slightly within $\text{dom } f$. We obtain entire maps g_i with arbitrarily small uniform error on the larger products and hence, by Cauchy estimates, arbitrarily small error in every prescribed finite number of derivatives on the retained products. On a neighborhood of $\overline{D}_b \times C \times P$, two adjacent maps g_i, g_{i+1} therefore lie in the fixed inverse fiber charts of a chart spray of f and can be written as

$$g_i = f^\sharp(\cdot, a_i), \quad g_{i+1} = f^\sharp(\cdot, a_{i+1}).$$

Choose a fixed smooth cutoff χ on $[0, 1]$ with $\chi = 0$ near 0 and $\chi = 1$ near 1, and set

$$(3.10) \quad f_\tau = f^\sharp(\cdot, (1 - \chi(\tau))a_i + \chi(\tau)a_{i+1}), \quad 0 \leq \tau \leq 1.$$

The family in (3.10) is defined near C , and equals the entire map g_i or g_{i+1} on the corresponding endpoint subintervals. For $v_\tau = (1 - \chi(\tau))a_i + \chi(\tau)a_{i+1}$,

$$(3.11) \quad \partial_\tau^\ell v_\tau = \chi^{(\ell)}(\tau)(a_{i+1} - a_i) \quad (\ell \geq 1), \quad \|v_\tau\|_{C^j} \leq C_{j,\chi}(\|a_i\|_{C^j} + \|a_{i+1}\|_{C^j}).$$

Since $f^\sharp(\cdot, 0) = f$, the chain rule, (3.11), and uniform bounds for the derivatives of the fixed spray on each retained compact piece K give, for sufficiently accurate choices of g_i and g_{i+1} ,

$$(3.12) \quad \|f_\tau - f\|_{C^j([0,1] \times K)} \leq C_j(\|a_i\|_{C^j(K)} + \|a_{i+1}\|_{C^j(K)}) \longrightarrow 0.$$

Paths on a common domain. Rescaling and entire approximation replace the local path by one available on the common larger polydisc B . Choose $d_* \in \text{int } C$ and

$$(3.13) \quad r_C = \text{dist}(d_*, \partial C) > 0, \quad R_B = \max_{y \in B} |y - d_*|, \quad L > \max(1, R_B/r_C).$$

Equation (3.13) gives

$$d_* + L^{-1}(B - d_*) \Subset \text{int } C.$$

Choose smooth functions $\kappa_0, \kappa_1 : [0, 1] \rightarrow [0, 1]$ with disjoint supports in the intervals where $f_\tau = g_i$ and $f_\tau = g_{i+1}$, respectively, and with $\kappa_0 = 1$ near 0, $\kappa_1 = 1$ near 1. The scaling function is

$$(3.14) \quad \lambda(\tau) = 1 + (L - 1)(\kappa_0(\tau) + \kappa_1(\tau)), \quad 1 \leq \lambda(\tau) \leq L.$$

Set

$$(3.15) \quad \tilde{f}_\tau(t, y, u) = f_\tau(t, d_* + \lambda(\tau)(y - d_*), u).$$

It is well defined near the common small set C : by (3.14), the condition $\lambda > 1$ forces f_τ to be one of the entire endpoint maps. Fix g_i, g_{i+1}, λ, B and chart sprays of both endpoint maps on a neighborhood of $D_b \times B \times P$ before choosing the next approximation.

Apply Lemma 3.1 to $\tilde{f}$ in (3.15), followed by Lemma 3.2 with A_{r-1} , treating the t -disc and u variables as passive. This gives a jointly entire approximation $E(\tau, t, y, u)$. Set

$$(3.16) \quad A_\tau(t, y, u) = E(\tau, t, d_* + \lambda(\tau)^{-1}(y - d_*), u).$$

Since E is entire, (3.16) makes A_τ available on all of B . For $y \in C^-$, set $y_\tau = d_* + \lambda(\tau)^{-1}(y - d_*)$. Then

$$(3.17) \quad y_\tau \in \text{conv}(C^- \cup \{d_*\}) \Subset \text{int } C, \quad \tilde{f}_\tau(t, y_\tau, u) = f_\tau(t, y, u).$$

For a closed disc $\overline{D}'_b \Subset D_b$ about the subdivision point, set $K_- = \overline{D}'_b \times C^- \times P$. Given a derivative order j , choose a compact product K_C in the approximation domain containing all (τ, t, y_τ, u) with $(t, y, u) \in K_-$. By (3.16), (3.17), and the chain rule,

$$(3.18) \quad \|A_\tau - f_\tau\|_{C^j([0,1] \times K_-)} \leq C_j \|E - \tilde{f}\|_{C^j(K_C)},$$

where C_j depends on the fixed scaling function and the retained products. The right side of (3.18) can be made arbitrarily small. Together with (3.12), this makes A_τ arbitrarily close to the original map f on the retained product. On smaller compact endpoint intervals E_0, E_1 , the scaling satisfies

$$(3.19) \quad \lambda(\tau) = L, \quad S_L(y) = d_* + L^{-1}(y - d_*), \quad S_L(B) \subseteq \text{int } C.$$

Writing $g^{(0)} = g_i$ and $g^{(1)} = g_{i+1}$, equations (3.15) and (3.19) give

$$(3.20) \quad \tilde{f}_\tau(t, S_L(y), u) = g^{(\ell)}(t, y, u), \quad \tau \in E_\ell, \quad y \in B, \quad \ell = 0, 1.$$

Choose K_C also to contain the compact products obtained from the retained endpoint domains by $(\tau, t, y, u) \mapsto (\tau, t, S_L(y), u)$. By (3.16) and (3.20),

$$(3.21) \quad \|A - g^{(\ell)}\|_{C^j(E_\ell \times \overline{D'_b} \times B \times P)} \leq C_{j,L} \|E - \tilde{f}\|_{C^j(K_C)} \longrightarrow 0.$$

Thus (3.21) places A_τ in the previously fixed inverse fiber charts of the endpoint sprays over B .

Exact endpoint repair. To concatenate the approximating paths, their endpoint values must agree exactly with g_i and g_{i+1} . On E_ℓ , the inverse fiber coordinates of the fixed endpoint spray give

$$A_\tau = (g^{(\ell)})^\#(\cdot, \beta_\tau^{(\ell)}), \quad \ell = 0, 1.$$

Choose smooth cutoffs $\vartheta_\ell : E_\ell \rightarrow [0, 1]$ equal to 0 near the path endpoint $\tau = \ell$ and to 1 near the other, interior endpoint of E_ℓ . The corrected path is

$$(3.22) \quad \hat{A}_\tau = \begin{cases} (g^{(\ell)})^\#(\cdot, \vartheta_\ell(\tau)\beta_\tau^{(\ell)}), & \tau \in E_\ell, \quad \ell = 0, 1, \\ A_\tau, & \tau \notin E_0 \cup E_1. \end{cases}$$

At the interior endpoint of each E_ℓ , equation (3.22) agrees with A_τ on a neighborhood, so the corrected path is smooth. At the endpoints of the full path interval it satisfies

$$\hat{A}_\tau = g_i \quad \text{for } \tau \text{ near } 0, \quad \hat{A}_\tau = g_{i+1} \quad \text{for } \tau \text{ near } 1.$$

The corrections are available over B by (3.21) and are supported where $\lambda = L$. The endpoint sprays, their inverse charts, and the cutoffs are fixed. The chain rule therefore gives, on each retained endpoint product,

$$\|\hat{A} - g^{(\ell)}\|_{C^j} \leq C_j \|A - g^{(\ell)}\|_{C^j} \longrightarrow 0, \quad \ell = 0, 1,$$

where the norms include derivatives in τ . Thus the repair preserves the required finite derivative accuracy before the path parameter is replaced by a function of t .

Assembly and complexification. Write the subdivision intervals as $I_i = [t_{i-1}, t_i]$. Around each inner division point t_i , choose disjoint joining intervals $J_i = (t_i - \delta_i, t_i + \delta_i) \subseteq D'_b \cap \mathbb{R}$ and fixed smooth functions $\rho_i : J_i \rightarrow [0, 1]$ equal to 0 near the left endpoint and 1 near the right endpoint. Denote the path just constructed between g_i and g_{i+1} by $\hat{A}^{(i)}$. With $I_i^0 = I_i \setminus \bigcup_\ell J_\ell$, the family on the common domain is

$$(3.23) \quad H(t, y, u) = \begin{cases} \hat{A}_{\rho_i(t)}^{(i)}(t, y, u), & t \in J_i, \\ g_i(t, y, u), & t \in I_i^0. \end{cases}$$

Near the ends of J_i , (3.22) makes the first expression in (3.23) equal to g_i and g_{i+1} , respectively. Thus H is smooth in t and holomorphic in (y, u) near $I \times B \times P$. For the compact pieces

$$\mathcal{K}_i = (F \cap (I_i \times \mathbb{C}^{r-1})) \times P, \quad \mathcal{J}_i = (F \cap (\overline{J}_i \times \mathbb{C}^{r-1})) \times P,$$

and any prescribed derivative order j , the chain rule and $\rho_i(J_i) \subset [0, 1]$ give

$$(3.24) \quad \|H - f\|_{C^j(F \times P)} \leq C_j \max_i \|g_i - f\|_{C^j(\mathcal{K}_i)} + C_j \max_i \|\hat{A}^{(i)} - f\|_{C^j([0,1] \times \mathcal{J}_i)},$$

where f is independent of the path parameter and C_j depends on the fixed ρ_i . The two errors in (3.24) can be made arbitrarily small, the first by the local entire approximation and the second by (3.12), (3.18), (3.21), and (3.22). Extend H slightly beyond the endpoints of I by its first and last entire maps. Apply Lemma 3.1 again to obtain H_1 , jointly holomorphic near a complex neighborhood of $I \times B \times P$. Since B is a polydisc, Lemma 3.2 applied to H_1 , with A_{r-1} and the u -variables as passive parameters, now gives an entire map in all variables. When $r = 1$, there is no y -variable and the rescaling step is omitted. $\square$

3.2. Real-interface gluing and convex approximation. To pass from approximation on real faces to convex approximation, we must turn two close traces on a real hyperplane into exact agreement while controlling the change on the old side. We first fix the norms in which this correction is performed.

For $b > 0$, let

$$B_b = \{v \in \mathbb{C}^n : |v| < b\}$$

denote the Euclidean ball in the fiber variables.

Fix $0 < \alpha < 1$. For a subset M of a Euclidean space and a vector-valued function u , write

$$(3.25) \quad \|u\|_{C^\alpha(M)} = \sup_M |u| + [u]_{\alpha, M}, \quad [u]_{\alpha, M} = \sup_{\substack{z, w \in M \\ z \neq w}} \frac{|u(z) - u(w)|}{|z - w|^\alpha}.$$

For $Q \subset \mathbb{C}^m$ compact and convex, write

$$(3.26) \quad Q_r = \{z \in \mathbb{C}^m : \text{dist}(z, Q) < r\}, \quad r > 0.$$

The interface and its two sides are

$$(3.27) \quad M_r = Q_r \cap \{\text{Im } z_1 = 0\}, \quad Q_r^\pm = Q_r \cap \{\pm \text{Im } z_1 > 0\}.$$

The norms used for traces and for the two sides are

$$(3.28) \quad \begin{aligned} \|u\|_{r, b, \alpha} &= \sup_{v \in B_b} \|u(\cdot, v)\|_{C^\alpha(M_r)}, \\ \|a\|_{r, b, *}^\pm &= \sup_{Q_r^\pm \times B_b} |a| + \sup_{v \in B_b} \|a(\cdot, v)\|_{M_r} \|_{C^\alpha(M_r)}. \end{aligned}$$

The difference quotient in (3.25) always uses all base variables. The fiber variable v is subject to a supremum and is treated by holomorphic estimates on smaller balls.

The additive splitting of [46, Lemma 4.4] plays the role of Lemma 2.5 in the full-base C^α norms (3.28). Its nonlinear correction gives the following real-interface gluing theorem [46, Lemma 4.5], the real-interface counterpart of [46, Lemma 3.2]. The corrections also control the change on the old side, allowing successive cuts with a prescribed total error.

Lemma 3.4. *For a compact convex $Q \subset \mathbb{C}^m$ with the geometry (3.26)–(3.27), fix $r, b > 0$, an outer fiber margin $\mu > 0$, losses $0 < \sigma < r/2$, $0 < \tau < b$, and $\eta > 0$. There exists $\delta \in (0, \mu)$ such that the following holds. Suppose that F_-, F_+ are holomorphic on $Q_r^- \times B_{b+\mu}$, $Q_r^+ \times B_b$, continuous up to their interface, and their traces satisfy on $M_r \times B_b$*

$$F_-(z, \gamma(z, v)) = F_+(z, v), \quad \gamma(z, v) = v + c(z, v), \quad \|c\|_{r, b, \alpha} < \delta.$$

The transition is holomorphic in the complex tangential variables and in v . Under these hypotheses there are holomorphic fiber reparametrizations $\alpha_\pm = \text{id} + u_\pm$ on $Q_{r-\sigma}^\pm \times B_{b-\tau}$, with C^α traces and values in B_b , such that

$$\begin{aligned} \|u_-\|_{r-\sigma, b-\tau, *}^- + \|u_+\|_{r-\sigma, b-\tau, *}^+ &< \eta, \\ F_-(z, \alpha_-(z, v)) &= F_+(z, \alpha_+(z, v)) \quad (z \in M_{r-\sigma}). \end{aligned}$$

The two corrected central maps join holomorphically across the interface. Denote the resulting core by

$$F(z) = \begin{cases} F_-(z, \alpha_-(z, 0)), & z \in Q_{r-\sigma}^- \cup M_{r-\sigma}, \\ F_+(z, \alpha_+(z, 0)), & z \in Q_{r-\sigma}^+ \cup M_{r-\sigma}. \end{cases}$$

If $K \Subset Q_{r-\sigma}^- \cup M_{r-\sigma}$, then

$$(3.29) \quad \sup_K d_Y(F, F_-(\cdot, 0)) \leq \omega_{-,K}(\eta),$$

where $\omega_{-,K}$ is a continuity modulus of the fixed spray F_- in its fiber variable.

Proof. The reparametrizations and their C^α estimates are the quantitative content of [46, Lemma 4.5], whose proof combines the full-base additive splitting of [46, Lemma 4.4] with the quadratic iteration of [46, Lemma 3.2]. Those arguments allow fiber variables of arbitrary finite dimension. For the fixed losses σ, τ and margin μ , their estimates tend to zero with $\|c\|_{r,b,\alpha}$; choosing δ smaller therefore gives the displayed bound by η . The transition identity makes the corrected central maps agree on the interface, so the cited gluing lemma joins them holomorphically. Finally,

$$d_Y(F_-(z, \alpha_-(z, 0)), F_-(z, 0)) \leq \omega_{-,K}(|u_-(z, 0)|) \leq \omega_{-,K}(\eta)$$

on K , which proves (3.29). $\square$

Proof of Theorem 2.3. We prove A_r by induction on r , following [46, Section 4.4, proof of Theorem 3.1]. The base case A_0 follows from Proposition 2.9. Assume that A_{r-1} holds.

Extension across one affine cut. For a nonconstant real affine function,

$$(3.30) \quad \ell(z) = \operatorname{Re} \sum_{j=1}^r a_j z_j + c, \quad w_1 = i \left(\sum_{j=1}^r a_j z_j + c - b \right), \quad \operatorname{Im} w_1 = \ell(z) - b.$$

Complete the nonzero linear part of w_1 to a basis of complex linear functionals and retain the translation in (3.30). After renaming the resulting affine coordinates,

$$Q_- = Q \cap \{\operatorname{Im} z_1 \leq 0\} \subset Q \subset \mathbb{C}^r,$$

where Q is compact and convex and f is holomorphic near $Q_- \times P$. When Q_- has nonempty interior and $Q_- \neq Q$, the cutting hyperplane meets $\operatorname{int} Q$; this is the case treated here. Take a slightly larger passive polydisc $P^+ \supseteq P$. For small $r_* > 0$,

$$(3.31) \quad (\overline{Q_{r_*}} \cap \{\operatorname{Im} z_1 \leq 0\}) \times P^+ \subset \operatorname{dom} f.$$

Indeed, otherwise points outside $\operatorname{dom} f$ with $\operatorname{Im} z_1 \leq 0$ and distance to $Q \times P^+$ tending to zero would have a subsequence converging to a point of $Q_- \times P^+$. This contradicts the compact containment $Q_- \times P^+ \Subset \operatorname{dom} f$.

Lemma 2.4, applied on the convex lower-side domain supplied by (3.31), gives a fixed chart spray $f^\sharp(z, u, v)$ with fiber domain Δ_β^n . Before approximating it, fix $b, \mu > 0$, $0 < \tau < b$, and $\beta_0 < \beta$ such that

$$\overline{B_{b+\mu}} \Subset \Delta_{\beta_0}^n \Subset \Delta_\beta^n.$$

Apply Lemma 3.3 on a slightly larger compact real face, with the reserved u -polydisc and $\Delta_{\beta_0}^n$ as passive factors. It gives an entire map $G(z, u, v)$ approximating the trace of $f^\sharp$ with tangential derivative control through order two. The order-two accuracy and the Cauchy estimates on the reserved parameter margins give uniform first-derivative control on the smaller products used below. On the interface, invert the fixed fiber charts of $f^\sharp$ to obtain a transition $\gamma = \operatorname{id} + c$; the fixed inverse charts and the chain rule give, on a retained face,

$$(3.32) \quad \|c\|_{C^1} \leq C \|G - f^\sharp\|_{C^1} \longrightarrow 0.$$

The norms in (3.32) are measured in the fixed charts and include all base variables (z, u) , while the derivatives in u follow from Cauchy estimates on the reserved parameter margin. Since the face is convex and has bounded diameter D_0 ,

$$(3.33) \quad \|c\|_{C^\alpha} \leq \|c\|_{C^0} + D_0^{1-\alpha} \|D_{z,u}c\|_{C^0} \longrightarrow 0.$$

For the gluing, the compact convex base and its complex dimension are

$$(3.34) \quad \mathcal{Q} = Q \times P, \quad m = r + \dim P,$$

with coordinates (z, u) ; only $v \in \mathbb{C}^n$ remains a fiber variable. Small Euclidean thickenings of $\mathcal{Q}$ lie in the reserved product neighborhoods, and the C^α norm of the transition includes differences in all (z, u) variables. By (3.32), (3.33), and the Cauchy estimates in u ,

$$(3.35) \quad \sup_{v \in B_b} \|c(\cdot, v)\|_{C^\alpha(\mathcal{Q}_{r_*} \cap \{\operatorname{Im} z_1 = 0\})} \longrightarrow 0.$$

Thus Lemma 3.4 applies on $\mathcal{Q}_{r_*}$ with the base (3.34), the fixed fiber margins above, and any prescribed loss $0 < \sigma < r_*/2$, by (3.35). Since the old chart spray is fiberwise injective, equality of the corrected traces is equivalent to the transition identity. The resulting corrections satisfy

$$(3.36) \quad \gamma(z, u, \alpha_+(z, u, v)) = \alpha_-(z, u, v) \quad \text{on the interface,} \quad \|\alpha_- - \operatorname{id}\|_* + \|\alpha_+ - \operatorname{id}\|_* < \eta.$$

The resulting map on $\mathcal{Q}_{r_*-\sigma}$ is

$$(3.37) \quad \tilde{f}(z, u) = \begin{cases} f^\sharp(z, u, \alpha_-(z, u, 0)), & \operatorname{Im} z_1 \leq 0, \\ G(z, u, \alpha_+(z, u, 0)), & \operatorname{Im} z_1 \geq 0. \end{cases}$$

The two traces in (3.37) agree by (3.36), so Lemma 3.4 makes $\tilde{f}$ holomorphic near $Q \times P$. Equation (3.29) gives

$$(3.38) \quad \sup_{Q_- \times P} d_Y(\tilde{f}, f) \leq \omega_{f^\sharp, Q_- \times P}(\eta) \longrightarrow 0 \quad (\eta \longrightarrow 0).$$

Successive removal of the faces. To obtain approximation on an arbitrary compact convex set $K \subset \mathbb{C}^r$, take a map f holomorphic near $K \times P$. After thickening K and P inside the original domain, choose a full-dimensional real convex polytope K' with

$$K \Subset \operatorname{int} K', \quad K' \times P \Subset \operatorname{dom} f.$$

Choose R with $K' \Subset \operatorname{int} \overline{\Delta}_R^r$ and real affine inequalities $\ell_j(z) \leq b_j$ such that

$$(3.39) \quad K' = \bigcap_{j=1}^M \{\ell_j \leq b_j\}, \quad Q_k = \overline{\Delta}_R^r \cap \bigcap_{j=k+1}^M \{\ell_j \leq b_j\} \quad (0 \leq k \leq M),$$

so that

$$(3.40) \quad Q_0 = K', \quad Q_M = \overline{\Delta}_R^r, \quad Q_{k-1} = Q_k \cap \{\ell_k \leq b_k\}.$$

Every Q_k in (3.39) is compact and convex and contains the full-dimensional set K' , and whenever a step enlarges the set, its cutting hyperplane meets $\operatorname{int} Q_k$. By (3.30), the cut in (3.40) reduces to the case treated above. Applying (3.38) successively, starting with $f_0 = f$, gives f_k holomorphic near $Q_k \times P$ with

$$(3.41) \quad \sup_{Q_{k-1} \times P} d_Y(f_k, f_{k-1}) < \epsilon_k, \quad \sum_{k=1}^M \epsilon_k < \epsilon/2.$$

Using (3.41), Proposition 2.9 gives an entire F with

$$\sup_{\overline{\Delta}_R^r \times P} d_Y(F, f_M) < \epsilon/2, \quad \sup_{K \times P} d_Y(F, f) < \epsilon.$$

This proves A_r . Setting $q = 0$ in the statement, so that no passive parameters remain, gives CAP. $\square$

The amplification theorem now combines with the prescribed-parameter bridge in Proposition 2.2.

Proof of Theorems 1.2 and 1.3. The zero-dimensional case is immediate. By Proposition 2.2, Theorem 2.3, and Definition 1.1, we have

$$\text{strong dominability} \implies \mathcal{L}_n \implies \text{CAP} \iff \text{Oka}.$$

Conversely, every Oka manifold is strongly dominable [26, Proposition 7.1.9(a)]. This completes both equivalences. $\square$

Proof of Corollary 1.4. Only the converse requires proof. For $n = \dim X$, the property CAP_n implies strong dominability [26, Section 7.1]; Theorem 1.3 now applies. $\square$

Proof of Corollary 1.5. Such a stratification makes X strongly dominable [29, Theorem 2]; apply Theorem 1.3. $\square$

Proof of Corollary 1.6. Each Oka neighborhood strongly dominates the ambient manifold at its points. Hence X is strongly dominable, and Theorem 1.3 applies. $\square$

4. EXTENSION ACROSS A PROPER ANALYTIC SUBSET

We prove the union theorem by propagating LEAP across a proper analytic subset. For surfaces, the closing step in the corresponding argument uses two transverse target directions [46, Corollary 2.10]. In dimension $n > 2$, two directions do not form local coordinates, and the approximating long discs produced by planar enlargement do not yet give families fixing scalar zero exactly. The proof below retains the coordinate-disc and planar-enlargement construction of [46], but replaces the closing step by n recentered scalar families and the scalar-propagation method of [4, Lemma 4.3 and Section 4.4]. The families are defined on scalar discs of increasing radius, while their convergence is required only near scalar zero.

4.1. Propagation of LEAP. For a complex manifold Y of dimension $n \geq 1$, let

$$\mathcal{G} = \{x \in Y : Y \text{ has property } \mathcal{L}_n \text{ at } x\}.$$

The LEAP locus $\mathcal{G}$ is open: every point of a witness neighborhood has that same witness neighborhood. We will show that $Y \setminus \mathcal{G}$ is empty if it is locally contained in a proper analytic subset.

In this section the fiber domains are the maximum-norm polydiscs

$$\Delta_r^n = \{w \in \mathbb{C}^n : \|w\|_\infty < r\};$$

vector norms are the maximum norm and matrix norms its induced operator norm. Closed polydiscs retain the notation $\overline{\Delta}_r^n$. All maps near compact products are restricted, when needed, to a product of an open planar neighborhood and an open polydisc. These sources satisfy the triviality hypothesis of Lemma 2.4. Its fiber-preserving coordinates and uniform inverse-chart margins will be used below.

The next lemma adapts the scalar-substitution argument of An–Xie [4, Lemma 4.3]. Their family is entire in the new scalar variable, whereas here only the approximating families have scalar radii tending to infinity and convergence is local near zero. Substitution of a holomorphic scalar function nevertheless preserves approximability on the original source set. The increasing radii are essential: a polynomial that approximates the scalar function on the original compact set can be arbitrarily large on the gluing side, so the polynomial is fixed first and its range is absorbed by choosing the family index afterward.

Lemma 4.1. *Fix a special planar disc pair $D \subset D'$ and a closed polydisc $P_* \subset \mathbb{C}^q$, where $q \geq 0$. Write*

$$K = D \times P_*, \quad L = D' \times P_*, \quad s = (z, p),$$

and denote by $\mathcal{A}$ the maps in $\mathcal{O}(\text{Op } K, Y)$ uniformly approximable on K by maps in $\mathcal{O}(\text{Op } L, Y)$.

Suppose $f_0 \in \mathcal{A}$, Ω is an open neighborhood of K , and

$$G : \Omega \times \Delta_r^n \rightarrow Y, \quad G(s, 0) = f_0(s),$$

is holomorphic, with $(s, w) \mapsto (s, G(s, w))$ biholomorphic onto its image. Suppose

$$T_\nu : \Omega \times \Delta_r^n \times \Delta_{R_\nu} \rightarrow Y, \quad T_\nu(s, w, 0) = G(s, w), \quad R_\nu \rightarrow \infty,$$

are holomorphic and converge locally uniformly on $\Omega \times \Delta_r^n \times \Delta_\tau$, for some $\tau > 0$, to a holomorphic map T . For every $\varphi \in \mathcal{O}(\text{Op } K, \mathbb{C})$ with $\|\varphi\|_K < \tau$, the map

$$s \mapsto T(s, 0, \varphi(s))$$

belongs to $\mathcal{A}$.

Proof. Initial approximation and planar domains. Fix $\epsilon > 0$ and numbers $\|\varphi\|_K < t_0 < t_1 < \tau$. Since $f_0 \in \mathcal{A}$, there is $g \in \mathcal{O}(\text{Op } L, Y)$ so close to f_0 on K that

$$\rho(s) = G(s, \cdot)^{-1}(g(s))$$

is holomorphic near K , $\|\rho\|_K < r/16$, and

$$(4.1) \quad \sup_K d_Y(T(s, \rho(s), \varphi(s)), T(s, 0, \varphi(s))) < \epsilon/4.$$

The last assertion follows from uniform continuity on a compact neighborhood of $K \times \{0\} \times \varphi(K)$. Lemma 2.4 supplies a family $F(s, v)$ centered at g near L , available for $v \in \Delta_{2b}^n$, with inverse fiber coordinates Ψ_s .

We specify the planar geometry used for the two families. Write $D' = D \cup E$, where E is the attached disc. In E , a crosscut close to $D \cap E$, but disjoint from D , separates a small collar of D from the rest of E . A narrow closed neighborhood of that crosscut and small thickenings of the two sides give compact Jordan regions A, B such that

$$D \subset \text{int } B, \quad D' \subset \text{int}(A \cup B), \quad C = A \cap B \text{ is a closed Jordan disc disjoint from } D, \\ \overline{A \setminus B} \cap \overline{B \setminus A} = \emptyset.$$

The collar can be as thin as needed, so $B \times P_*$ lies in the common domain of G, ρ, φ , and F is available near $A \times P_*$. There are disjoint closed Jordan neighborhoods D^+, C^+ of D, C , a compact neighborhood B^+ of B , and polydiscs

$$P_* \Subset \text{int } P^+ \Subset \text{int } P^{++}$$

such that

$$D \Subset \text{int } D^+, \quad C \Subset \text{int } C^+, \quad D^+ \cup C^+ \subset B^+, \quad B \Subset \text{int } B^+,$$

and $B^+ \times P^{++}$ stays in those common domains, with $\|\rho\| < r/8$ there. The family F is available near both $A \times P^{++}$ and $C^+ \times P^{++}$. All these choices precede the index ν and all polynomial approximations below.

Choose fixed open planar neighborhoods U_A, U_B of A, B that cover a neighborhood of D' and have separated fringes. They can be taken with

$$U_A \cap U_B \Subset \text{int } C^+, \quad U_B \Subset \text{int } B^+,$$

and within the relevant family domains. A fixed open polydisc U_P satisfies $P_* \Subset U_P \Subset \text{int } P^+$. Lemma 2.5 applies with a fixed constant C_0 .

The sets $C^+ \times P^{++}$ and $(D^+ \cup C^+) \times P^+$ are polynomially convex, since the planar factors have connected complements and polynomial hulls commute with products. This allows a matrix approximation on the overlap and a scalar approximation with different data near D and C .

Uniform estimates and choice of parameters. On a neighborhood of $(s, \rho(s), 0)$ over $C^+ \times P^{++}$, consider

$$\mathcal{H}_\nu(s, w, u) = \Psi_s(T_\nu(s, w, u)).$$

The exact normalization at scalar zero gives

$$(4.2) \quad \mathcal{H}_\nu(s, \rho(s), 0) = 0, \quad \mathcal{E}(s) := \partial_w \mathcal{H}_\nu(s, \rho(s), 0) = \partial_w(\Psi_s \circ G)(s, \rho(s)) \in \mathrm{GL}_n(\mathbb{C}).$$

In particular, both the center and $\mathcal{E}$ are independent of ν . The inverse chart of F contains a fixed neighborhood of the compact graph of g on this product. Since $T(s, \rho(s), 0) = g(s)$, continuity first gives a common smaller neighborhood in (w, u) for the limit. Local uniform convergence then places all sufficiently large T_ν in the same inverse chart there. On still smaller compact products, Cauchy estimates give uniform bounds for the first and second derivatives of $\mathcal{H}_\nu$. Likewise, on a fixed compact neighborhood of

$$K \times \overline{\Delta_{r/2}^n} \times \overline{\Delta_{t_1}},$$

the first derivatives of T_ν in w, u are uniformly bounded in fixed finitely many target charts containing compact pieces of the limit image with positive margins. The source compact sets and target charts are independent of ν , and the same charts contain the corresponding images of all sufficiently large T_ν .

Write

$$N = 1 + \|\mathcal{E}^{-1}\|_{C^+ \times P^{++}}.$$

Fix a transition tolerance $\varepsilon_{\mathrm{tr}} > 0$ so small that a transition within $\varepsilon_{\mathrm{tr}}$ of the identity on Δ_2^n satisfies (2.3) on Δ_1^n and gives corrections of norm less than $1/4$. Here the derivative bound follows from Cauchy estimates, and the splitting constant C_0 is already fixed. The uniform Taylor bounds above give the estimate (4.7) below with a constant $c > 0$ depending only on the fixed products, $\mathcal{E}$, N , and those derivative bounds. In particular, c is independent of the polynomial matrix chosen below and of its size on $B^+ \times P^+$.

Choose $0 < \eta \leq 1$ with $c\eta < \varepsilon_{\mathrm{tr}}/3$. Oka–Weil approximation [19, Theorem 18] and Taylor truncation give a polynomial matrix L_B such that

$$\|L_B - \mathcal{E}^{-1}\|_{C^+ \times P^{++}} < \eta, \quad \|L_B\|_{C^+ \times P^{++}} \leq N.$$

Fix this matrix and set

$$M = \max\{1, \|L_B\|_{B^+ \times P^+}\} < \infty.$$

The derivative bounds on the fixed target charts and uniform continuity of their inverse coordinate maps give moduli ω_w, ω_u , independent of all sufficiently large ν , such that

$$(4.3) \quad \begin{aligned} d_Y(T_\nu(s, w, u), T_\nu(s, w', u)) &\leq \omega_w(\|w - w'\|_\infty), \\ d_Y(T_\nu(s, w, u), T_\nu(s, w, u')) &\leq \omega_u(|u - u'|) \end{aligned}$$

on $K \times \overline{\Delta_{r/2}^n} \times \overline{\Delta_{t_1}}$, with $\omega_w(h), \omega_u(h) \rightarrow 0$ as $h \rightarrow 0$. Choose $\delta > 0$ sufficiently small that the uniform overlap Taylor estimates apply, $c\delta < \varepsilon_{\mathrm{tr}}/3$, and

$$(4.4) \quad \delta < b/2, \quad \delta M < r/8, \quad \omega_w(\delta M/4) < \epsilon/4.$$

Choose $e_0 > 0$ within the fixed scalar margin for the overlap Taylor estimates and so small that

$$ce_0/\delta < \varepsilon_{\mathrm{tr}}/3.$$

Choose $e_1 > 0$ such that

$$(4.5) \quad e_1 < t_0 - \|\varphi\|_K, \quad \omega_u(e_1) < \epsilon/4.$$

With these errors fixed, the Oka–Weil theorem [19, Theorem 18] gives a scalar polynomial u_B satisfying

$$(4.6) \quad \|u_B - \varphi\|_{D^+ \times P^+} < e_1, \quad \|u_B\|_{C^+ \times P^+} < e_0.$$

Indeed, the data φ and 0 are holomorphic on disjoint neighborhoods of the two polynomially convex pieces. Equations (4.6) and (4.5) give $|u_B| < t_0$ on K .

Transition estimate and the family index. On the two sides consider the families

$$F_A(s, \zeta) = F(s, \delta\zeta), \quad F_{B,\nu}(s, \zeta) = T_\nu(s, \rho(s) + \delta L_B(s)\zeta, u_B(s)).$$

The first is available for $\zeta \in \Delta_4^n$. On the second side,

$$\|\rho(s) + \delta L_B(s)\zeta\| < r/8 + 2\delta M < r/2 \quad (\zeta \in \Delta_2^n).$$

On the overlap its transition is

$$\gamma_\nu(s, \zeta) = \delta^{-1} \Psi_s(F_{B,\nu}(s, \zeta)).$$

Taylor expansion at $(w, u) = (\rho(s), 0)$, using (4.2), gives

$$(4.7) \quad \|\gamma_\nu - \text{id}\|_{(U_A \cap U_B) \times U_P \times \Delta_2^n} \leq c(\eta + \delta + e_0/\delta),$$

uniformly for large ν . Indeed, (4.2) gives the exact expansion

$$(4.8) \quad \begin{aligned} \mathcal{H}_\nu(s, \rho(s) + \delta L_B(s)\zeta, u_B(s)) &= \delta \mathcal{E}(s) L_B(s)\zeta + \mathcal{R}_\nu(s, \zeta), \\ \|\mathcal{R}_\nu\|_\infty &\leq C(\delta^2 N^2 + e_0), \quad \|\mathcal{E} L_B - I\|_\infty \leq \|\mathcal{E}\|_\infty \eta. \end{aligned}$$

The quadratic term in (4.8) uses the uniform second derivative bound in w , and the term Ce_0 uses the first derivative bound in u and (4.6). Dividing (4.8) by δ and using $\|\zeta\|_\infty < 2$ gives (4.7), with c depending on the fixed products, N , and the uniform derivative bounds.

By the preceding choices, the right side of (4.7) is less than ε_{tr} . Cauchy estimates from Δ_2^n therefore give (2.3) on Δ_1^n , with corrections of norm less than $1/4$.

The increasing scalar radii enter here: the polynomial u_B is already fixed and has a finite norm on $B^+ \times P^+$. Choose ν last, so large that

$$R_\nu > 1 + \|u_B\|_{B^+ \times P^+}$$

and

$$(4.9) \quad \sup_K d_Y(T_\nu(s, \rho(s), \varphi(s)), T(s, \rho(s), \varphi(s))) < \epsilon/4.$$

Thus $u_B(B^+ \times P^+) \subset \Delta_{R_\nu}$, so $F_{B,\nu}$ is available on the full fixed B side. On the overlap, (4.6) keeps u_B in the fixed small scalar disc on which (4.8) holds.

Gluing and the approximation error. Applying Lemma 2.6 to the transition above gives corrections a, b on the two sides with

$$F_A(s, a(s)) = F_{B,\nu}(s, b(s)), \quad \|a\|, \|b\| < 1/4.$$

These expressions give a map $\tilde{g}$ holomorphic near L . On $K \subset B \times P_*$ it is

$$\tilde{g}(s) = T_\nu(s, \rho(s) + \delta L_B(s)b(s), u_B(s)).$$

For $s \in K$, the triangle inequality gives

$$\begin{aligned} d_Y(\tilde{g}(s), T(s, 0, \varphi(s))) &\leq d_Y(\tilde{g}(s), T_\nu(s, \rho(s), u_B(s))) \\ &\quad + d_Y(T_\nu(s, \rho(s), u_B(s)), T_\nu(s, \rho(s), \varphi(s))) \\ &\quad + d_Y(T_\nu(s, \rho(s), \varphi(s)), T(s, \rho(s), \varphi(s))) \\ &\quad + d_Y(T(s, \rho(s), \varphi(s)), T(s, 0, \varphi(s))) \\ &\leq \omega_w(\delta M/4) + \omega_u(e_1) + \epsilon/4 + \epsilon/4 < \epsilon. \end{aligned}$$

The four bounds follow from (4.3), (4.4), (4.5), (4.9), and (4.1). Since $\tilde{g} \in \mathcal{O}(\text{Op}(D' \times P_*), Y)$, the map $T(\cdot, 0, \varphi)$ belongs to $\mathcal{A}$. $\square$

Scalar substitution can now be iterated in independent tangent directions. This composition step follows the exact-intermediate-core argument of [4, Section 4.4]; the proposition below isolates the local target version needed here. Composing the limiting families gives local coordinates at x , so a map with image in a sufficiently small neighborhood of x can be obtained from the constant map by successive substitutions.

Proposition 4.2. *Let $x \in Y$, $\dim Y = n \geq 1$. Suppose on one neighborhood $V \ni x$ there are holomorphic maps*

$$h_{j,\nu} : V \times \Delta_{R_\nu} \rightarrow Y, \quad h_{j,\nu}(y, 0) = y, \quad 1 \leq j \leq n, \quad R_\nu \rightarrow \infty.$$

Suppose these maps converge locally uniformly on $V \times \Delta_\tau$, for some $\tau > 0$, to holomorphic maps h_j , and that $\partial_t h_j(x, 0)$, $1 \leq j \leq n$, form a basis of $T_x Y$. Then $x \in \mathcal{G}$.

Proof. The limits satisfy $h_j(y, 0) = y$. For $a = (a_1, \dots, a_n)$ near zero, consider the exact compositions

$$\theta_0(a) = x, \quad \theta_j(a) = h_j(\theta_{j-1}(a), a_j), \quad \Theta(a) = \theta_n(a).$$

On a sufficiently small polydisc $Q \ni 0$, every intermediate value belongs to a fixed relatively compact subset of V , and $|a_j| < \tau$. Since $d_y h_j(y, 0) = \text{id}$, the columns of $d\Theta_0$ are $\partial_t h_j(x, 0)$. The inverse function theorem gives a fixed neighborhood $W \ni x$ on which Θ^{-1} is available with image relatively compact in Q .

For arbitrary data $f \in \mathcal{O}(\text{Op}(D \times P), W)$ in the definition of $\mathcal{L}_n$, choose

$$P_0 \Subset \text{int } P_* \Subset \text{int } P, \quad (a_1, \dots, a_n) = \Theta^{-1} \circ f,$$

and consider near $K = D \times P_*$ the maps

$$f_0 = x, \quad f_j = h_j(f_{j-1}, a_j), \quad 1 \leq j \leq n.$$

They are exact holomorphic maps, $f_n = f$, and all their values lie in V ; also $\|a_j\|_K < \tau$. The constant map f_0 is in the approximation class $\mathcal{A}$ of Lemma 4.1.

If $f_{j-1} \in \mathcal{A}$, Lemma 2.4 gives coordinates $G_j(s, w)$ centered at f_{j-1} , with their whole image in V after restriction of the fixed base neighborhood and the fiber radius. The families

$$T_{j,\nu}(s, w, t) = h_{j,\nu}(G_j(s, w), t), \quad T_j(s, w, t) = h_j(G_j(s, w), t)$$

satisfy every hypothesis of Lemma 4.1; in particular their scalar-zero value is exactly $G_j(s, w)$. With $\varphi = a_j$, that lemma yields $f_j \in \mathcal{A}$. Finite induction gives $f \in \mathcal{A}$, and restriction to P_0 proves $\mathcal{L}_n$ at x . □

We obtain the required families from a coordinate disc whose boundary lies in $\mathcal{G}$. Lemma 2.7 enlarges the scalar domain; inverse branches then recenter the approximants so that scalar zero gives the identity exactly on one fixed target neighborhood. Repeated application of Lemma 4.1 requires this exact normalization, since convergence to the identity at scalar zero would not supply the induction hypothesis. For $n = 2$, the conclusion below is the first assertion of [46, Corollary 2.10]; the present argument derives it in arbitrary dimension from the scalar-family criterion above.

Proposition 4.3. *Let $\vartheta : U \rightarrow \Omega \subset \mathbb{C}^n$ be a coordinate chart, $x \in U$, and $v \neq 0$. Suppose for some $r > 0$ that*

$$\{\vartheta(x) + \zeta v : |\zeta| \leq r\} \subset \Omega, \quad \vartheta^{-1}(\{\vartheta(x) + \zeta v : |\zeta| = r\}) \subset \mathcal{G}.$$

Then $x \in \mathcal{G}$.

Proof. Independent directions and transverse discs. The closed disc is compactly contained in the chart, and its boundary is a compact subset of the open set $\mathcal{G}$. Hence all sufficiently small changes of the direction v preserve both properties. Choose n such directions $v_1, \dots, v_n$

which are linearly independent. Their choice and nonzero determinant are fixed before any approximation.

For each j , complete v_j to a basis $(v_j, e_{j,2}, \dots, e_{j,n})$ of $\mathbb{C}^n$ and consider

$$\psi_j(z, p) = \vartheta^{-1} \left(\vartheta(x) + zv_j + \sum_{k=2}^n p_{k-1} e_{j,k} \right).$$

A sufficiently small closed polydisc $P_j \subset \mathbb{C}^{n-1}$ makes this a coordinate parametrization near $\overline{\Delta}_r^1 \times P_j$. Moreover, for each $\zeta \in \partial\Delta_r$, the whole fiber $\psi_j(\{\zeta\} \times P_j)$ lies in a coordinate witness neighborhood for $\mathcal{L}_n$. To obtain one P_j for all ζ , first cover the boundary of the central disc by finitely many witness neighborhoods, use continuity to obtain corresponding source arcs, and then shrink the transverse polydisc so that the compact arc products stay in those neighborhoods.

For each j , choose $P'_j \Subset \text{int } P_j$ centered at zero. Fix one sequence $S_\nu > r$ with $S_\nu \rightarrow \infty$, to be used for all j . Applying Lemma 2.7 for each j and ν , with errors tending to zero, gives

$$\sigma_{j,\nu} \in \mathcal{O}(\text{Op}(\overline{\Delta}_{S_\nu}^1 \times P'_j), Y), \quad \sigma_{j,\nu} \longrightarrow \psi_j \quad \text{uniformly on } \overline{\Delta}_r^1 \times P'_j.$$

Recentering the approximants. On a smaller fixed product around $(0, 0)$, the images lie in the target chart, and Cauchy estimates give convergence of the first derivatives. Choose $\varrho_j > 0$ with $\overline{\Delta}_{\varrho_j}^n \Subset \Delta_r \times \text{int } P'_j$. In the coordinates of ψ_j , write

$$(4.10) \quad A_{j,\nu} = \psi_j^{-1} \circ \sigma_{j,\nu} = \text{id} + e_{j,\nu}, \quad \|e_{j,\nu}\|_{\overline{\Delta}_{\varrho_j}^n} < \varrho_j/4, \quad \|De_{j,\nu}\|_{\overline{\Delta}_{\varrho_j}^n} < 1/2$$

for all sufficiently large ν . Since there are only finitely many j , we discard one finite initial segment so that these estimates hold for every j . For $b \in \Delta_{\varrho_j/2}^n$, the map $a \mapsto b - e_{j,\nu}(a)$ satisfies, by (4.10),

$$\|b - e_{j,\nu}(a)\|_\infty < 3\varrho_j/4, \quad \|e_{j,\nu}(a) - e_{j,\nu}(a')\|_\infty \leq \frac{1}{2} \|a - a'\|_\infty.$$

It is therefore a contraction of $\overline{\Delta}_{\varrho_j}^n$ into its interior. Its unique fixed point $a_{j,\nu}(b)$ satisfies

$$A_{j,\nu}(a_{j,\nu}(b)) = b, \quad \|a_{j,\nu}(b) - b\|_\infty \leq \|e_{j,\nu}\|_{\overline{\Delta}_{\varrho_j}^n} \longrightarrow 0.$$

The invertibility of $DA_{j,\nu}$ in (4.10) makes $a_{j,\nu}$ holomorphic. On the fixed neighborhood

$$V = \bigcap_{j=1}^n \psi_j(\Delta_{\varrho_j/2}^n)$$

of x , this gives inverse branches

$$\ell_{j,\nu} = a_{j,\nu} \circ \psi_j^{-1} = (s_{j,\nu}, p_{j,\nu}), \quad \sigma_{j,\nu} \circ \ell_{j,\nu} = \text{id}_V.$$

Limiting scalar families. After another fixed restriction of V , the images of all these inverse branches lie in compact subproducts of $\Delta_r \times \text{int } P'_j$. They converge to ψ_j^{-1} . Since j ranges over a finite set, there is $C > 0$ such that $|s_{j,\nu}(y)| < C$ for every j , every $y \in V$, and every retained index. The radii $R_\nu = S_\nu - C$ tend to infinity; discard finitely many initial terms so that $R_\nu > 0$. The maps

$$h_{j,\nu}(y, t) = \sigma_{j,\nu}(s_{j,\nu}(y) + t, p_{j,\nu}(y))$$

are consequently holomorphic on the common product $V \times \Delta_{R_\nu}$ and satisfy

$$h_{j,\nu}(y, 0) = y$$

exactly. The radii R_ν tend to infinity.

There is one $\tau > 0$ such that, for $y \in V$ and $|t| < \tau$, all translated source points remain in a fixed compact subproduct of the original convergence product. Thus these maps converge locally uniformly on $V \times \Delta_\tau$ to

$$h_j(y, t) = \psi_j(\psi_j^{-1}(y) + (t, 0)).$$

Their derivatives $\partial_t h_j(x, 0)$ are $(d\vartheta_x)^{-1}v_j$ and form a basis. Proposition 4.2 gives $x \in \mathcal{G}$. $\square$

A suitable coordinate line through a point of a proper analytic subset avoids the subset on a small punctured disc, providing the boundary condition in Proposition 4.3. The following is the arbitrary-dimensional counterpart of the second assertion of [46, Corollary 2.10].

Corollary 4.4. *Suppose every point of Y has a connected neighborhood on which $Y \setminus \mathcal{G}$ is contained in a proper complex analytic subset. Then $\mathcal{G} = Y$.*

Proof. Consider a point $x \notin \mathcal{G}$. In a connected coordinate neighborhood centered at x , let B be a proper analytic subset containing the local complement of $\mathcal{G}$. Its ideal has a nonzero holomorphic germ a at x . In centered coordinates, write its first nonzero homogeneous term as a_m . Choose $v \in \mathbb{C}^n$ with $a_m(v) \neq 0$. Then

$$a(\zeta v) = \zeta^m a_m(v) + O(\zeta^{m+1})$$

is not identically zero. On a sufficiently small punctured disc it has no zeros. The corresponding closed coordinate-line disc lies in the chart and its boundary avoids B , hence lies in $\mathcal{G}$. Proposition 4.3 gives $x \in \mathcal{G}$, a contradiction. $\square$

The following consequence uses entire maps into the ambient manifold; their images need not avoid the analytic subset.

Corollary 4.5. *Let X be a complex manifold and $A \subset X$ a proper closed complex analytic subset. If X is dominable at every point of $X \setminus A$, then X is Oka.*

Proof. The case $\dim X = 0$ is immediate. For $n = \dim X \geq 1$, Proposition 2.2 gives

$$X \setminus A \subset \mathcal{G}, \quad X \setminus \mathcal{G} \subset A.$$

Since X is connected and A is proper, the germ of A is proper at every point. Corollary 4.4 yields $\mathcal{G} = X$, and Theorem 2.3 gives CAP. Thus X is Oka by Definition 1.1. $\square$

Proof of Theorem 1.7. By Theorem 1.3, the Oka manifold $X \setminus A$ strongly dominates X at every point of $X \setminus A$. Corollary 4.5 applies. $\square$

4.2. A criterion using entire-curve families. The geometric applications use the following special case of Proposition 4.2.

Definition 4.6. An n -dimensional complex manifold Y admits *local families of entire curves spanning the tangent spaces* at $x \in Y$ if there are an open neighborhood $V \ni x$, an integer $N \geq 1$, and holomorphic maps

$$h_j : V \times \mathbb{C} \longrightarrow Y, \quad h_j(y, 0) = y, \quad 1 \leq j \leq N,$$

such that $\partial_t h_1(y, 0), \dots, \partial_t h_N(y, 0)$ span $T_y Y$ for every $y \in V$. Each family is defined on the full product $V \times \mathbb{C}$; its image need not remain in V .

Corollary 4.7 (Entire-curve families). *A complex manifold Y satisfying the condition in Definition 4.6 at every point has CAP and is Oka. More generally, the same conclusion holds if this condition, with target Y , holds at every point of $Y \setminus A$, where $A \subset Y$ is a proper closed complex analytic subset. The images of the families may meet A . For a disconnected manifold the assertion holds componentwise, provided A is proper in each connected component.*

Proof. The zero-dimensional case is immediate. At a point x with the stated families, choose $n = \dim Y$ of their tangent vectors that form a basis of $T_x Y$, and relabel the corresponding families $h_1, \dots, h_n$. Their determinant in local coordinates is nonzero at x , so these vectors remain independent throughout a smaller neighborhood V . The restrictions $h_{j,\nu} = h_j|_{V \times \Delta_\nu}$ have radii tending to infinity, satisfy $h_{j,\nu}(y, 0) = y$, and agree with h_j on $V \times \Delta_1$. Proposition 4.2 therefore gives $\mathcal{L}_n$ at x . If the condition holds outside A , Corollary 4.4 gives $\mathcal{L}_n$ everywhere, applied on each connected component when necessary. Theorem 2.3 yields CAP. $\square$

5. RATIONALLY CONNECTED AND FANO MANIFOLDS

Deformations of very free rational curves give the families required by Corollary 4.7 at every point of a rationally connected projective manifold, and these families are holomorphic on the whole product $V \times \mathbb{P}^1$.

A nonconstant holomorphic map $u : \mathbb{P}^1 \rightarrow X$ is *very free* if u^*TX is ample; see [10, Definition 2.25]. By the Birkhoff–Grothendieck splitting theorem this means that

$$u^*TX \simeq \bigoplus_{\nu=1}^{\dim X} \mathcal{O}_{\mathbb{P}^1}(a_\nu), \quad a_\nu \geq 1;$$

see [39, Definition 3.1].

Proof of Theorem 1.8. Let X be a smooth complex projective rationally connected manifold. The zero-dimensional case is immediate. Fix $x \in X$, set $n = \dim X \geq 1$, and choose a basis $e_1, \dots, e_n$ of $T_x X$. Write t for the standard affine coordinate on $\mathbb{P}^1 \setminus \{\infty\} = \mathbb{C}$. The theorem on very free rational curves with prescribed tangent directions [11, Theorem 2.2(ii)] applies to X . Its single-point case, applied to the lines $\mathbb{C}e_j$ separately, gives holomorphic maps

$$u_j : \mathbb{P}^1 \longrightarrow X, \quad u_j(0) = x, \quad (du_j)_0 \left(\frac{\partial}{\partial t} \Big|_0 \right) \in \mathbb{C}^* e_j, \quad u_j^*TX \text{ ample}.$$

Here a source reparametrization sends the prescribed point to 0. Write $u_j^*TX \simeq \bigoplus_{\nu=1}^n \mathcal{O}_{\mathbb{P}^1}(a_{j\nu})$ with $a_{j\nu} \geq 1$. Let $[0]$ denote the divisor consisting of the point $0 \in \mathbb{P}^1$. Thus

$$H^1(\mathbb{P}^1, u_j^*TX) = 0, \quad H^1(\mathbb{P}^1, u_j^*TX \otimes \mathcal{O}_{\mathbb{P}^1}(-[0])) = 0,$$

since the corresponding line-bundle summands have degrees $a_{j\nu}$ and $a_{j\nu} - 1$, respectively.

We use the local deformation theory of maps with the source $\mathbb{P}^1$ fixed. Applied in local analytic form, the deformation results of [10, §1.4, Proposition 1.6 and Theorem 1.9] and the vanishing $H^1(\mathbb{P}^1, u_j^*TX) = 0$ give, for each j , a complex manifold S_j , a point $p_j \in S_j$, and a holomorphic family

$$F_j : S_j \times \mathbb{P}^1 \longrightarrow X, \quad F_j(p_j, t) = u_j(t).$$

The parameter derivative identifies $T_{p_j}S_j$ with $H^0(\mathbb{P}^1, u_j^*TX)$. Under this identification, the differential of evaluation $\text{ev}_{0,j} : S_j \rightarrow X$, $s \mapsto F_j(s, 0)$, is $\xi \mapsto \xi(0)$. The evaluation exact sequence and the vanishing $H^1(\mathbb{P}^1, u_j^*TX \otimes \mathcal{O}_{\mathbb{P}^1}(-[0])) = 0$ make this differential surjective. Hence $\text{ev}_{0,j}$ is a holomorphic submersion near p_j and has a local holomorphic section through p_j . Taking a common neighborhood V of x for the finitely many sections gives

$$\sigma_j : V \longrightarrow S_j, \quad \text{ev}_{0,j} \circ \sigma_j = \text{id}_V, \quad \sigma_j(x) = p_j.$$

We therefore obtain holomorphic maps

$$h_j : V \times \mathbb{P}^1 \longrightarrow X, \quad h_j(y, t) = F_j(\sigma_j(y), t).$$

They satisfy $h_j(y, 0) = y$ and $h_j(x, t) = u_j(t)$. Their derivatives at $(x, 0)$ lie in the lines $\mathbb{C}e_j$ and are nonzero. By continuity, after shrinking V we have

$$\frac{\partial h_1}{\partial t}(y, 0) \wedge \dots \wedge \frac{\partial h_n}{\partial t}(y, 0) \neq 0 \quad (y \in V).$$

Consequently these vectors span $T_y X$ for every $y \in V$. Restricting the second variable to $\mathbb{C} = \mathbb{P}^1 \setminus \{\infty\}$ gives the maps required in Definition 4.6. Corollary 4.7 therefore shows that X is Oka. Every smooth complex Fano manifold is rationally connected by Kollár–Miyaoka–Mori [40, Theorem 0.1], which gives the Fano case. $\square$

6. LOGARITHMIC RATIONAL CURVES AND DIVISOR COMPLEMENTS

6.1. A criterion using logarithmic rational curves. Let X be a connected complex projective manifold of dimension n and let $D = \sum_i D_i$ be a reduced divisor with irreducible components D_i . The divisor has *simple normal crossings* if each D_i is smooth and D is a union of coordinate hyperplanes in suitable local holomorphic coordinates. For a simple normal crossings divisor D , the holomorphic sections of the logarithmic tangent bundle $T_X(-\log D)$ are the holomorphic vector fields tangent to every component D_i . In coordinates $x_1, \dots, x_n$ with $D = \{x_1 \cdots x_k = 0\}$ this bundle has the frame

$$x_1 \frac{\partial}{\partial x_1}, \dots, x_k \frac{\partial}{\partial x_k}, \frac{\partial}{\partial x_{k+1}}, \dots, \frac{\partial}{\partial x_n}.$$

Theorem 6.1. *Let X be a connected complex projective manifold, and let $D \subset X$ be a reduced simple normal crossings divisor. Suppose that a nonconstant holomorphic map $f : \mathbb{P}^1 \rightarrow X$ satisfies*

$$f^{-1}(D) \subseteq \{\infty\}, \quad f^*T_X(-\log D) \text{ is ample.}$$

Then $X \setminus D$ is Oka.

We prove the theorem by deforming rational curves while keeping the pullback divisors fixed, so that their restrictions to $\mathbb{C}$ avoid the boundary. The logarithmic tangent bundle controls the values and first derivatives of the deformed curves.

Throughout this section, $D = \sum_{i=1}^r D_i$ has simple normal crossings, $U = X \setminus D$, and $n = \dim X$. We allow $r = 0$, in which case $D = 0$. Fix distinct points $0, \infty \in \mathbb{P}^1$ and an affine coordinate t on $\mathbb{P}^1 \setminus \{\infty\}$ with $t(0) = 0$. We write $g'(0) = dg_0(\partial/\partial t|_0) \in T_{g(0)}X$, and $[0]$ and $[\infty]$ denote the corresponding point divisors.

6.2. Prescribed pullback divisors and first jets. Fix f as in Theorem 6.1 and write $f^*D_i = m_i[\infty]$ with $m_i \geq 0$. We work with parametrized maps, keeping the source $\mathbb{P}^1$, the points $0, \infty$, and the coordinate t fixed. Fix a projective embedding of X and the degree $\deg f^*\mathcal{O}_X(1)$; the corresponding scheme of morphisms $\mathbb{P}^1 \rightarrow X$ is of finite type [10, §1.3]. Restrict it to the open-and-closed locus on which $\deg g^*\mathcal{O}_X(D_i) = m_i$ for all i , and let H be the open subset of maps whose image is contained in no D_i . Pulling back the divisors gives an algebraic morphism

$$c : H \longrightarrow \prod_{i=1}^r \text{Sym}^{m_i}(\mathbb{P}^1), \quad c(g) = (g^*D_i)_i.$$

Here $\text{Sym}^{m_i}(\mathbb{P}^1)$ parametrizes effective divisors of degree m_i , equivalently unordered m_i -tuples of points counted with multiplicity, so c records the intersection points with each D_i and their orders of contact. By the choice of H , the pulled-back defining section of D_i is not identically zero on any $\mathbb{P}^1$ -fiber. Its zero scheme is locally principal, and its restriction to each fiber is an effective Cartier divisor. Since $H \times \mathbb{P}^1 \rightarrow H$ is flat and of finite presentation, the fiberwise criterion [49, Tag 062Y, part (3)] shows that this zero scheme is a relative effective Cartier divisor, flat over H . It is thus a family of effective divisors of degree m_i and determines a morphism to $\text{Sym}^{m_i}(\mathbb{P}^1)$. We take $\text{Sym}^0(\mathbb{P}^1)$ and the empty product to be a point. Let

$$M = c^{-1}((m_i[\infty])_i)$$

be the scheme-theoretic fiber. M therefore parametrizes maps g with $g^*D_i = m_i[\infty]$ for every i . When all $m_i = 0$, we have $M = H$ and $g^*T_X(-\log D) \simeq g^*T_X$ for every $g \in M$; in particular, the case $D = \emptyset$ recovers the ordinary very free curve calculation. Every $g \in M$ satisfies $g(\mathbb{A}^1) \subset U$, so the universal evaluation map restricts to a morphism

$$(6.1) \quad \mathcal{F} : M \times \mathbb{A}^1 \longrightarrow U.$$

Lemma 6.2. *Let $g \in M$ be a map for which $E_g = g^*T_X(-\log D)$ is ample. Then the following hold.*

- (i) The space M is smooth at g , with tangent space $T_g M \simeq H^0(\mathbb{P}^1, E_g)$.
- (ii) The first-jet map

$$J : M \longrightarrow TU, \quad g \longmapsto (g(0), g'(0))$$

is smooth at g ; viewed as a holomorphic map near g , it is a submersion there. In particular, evaluation at 0 is smooth at g . On the fiber of evaluation at 0 over $g(0)$, derivative evaluation $\tilde{g} \mapsto \tilde{g}'(0) \in T_{g(0)}U$ has surjective differential at g .

Proof. Fixed-contact deformation space. The inclusion of the logarithmic vector fields in all vector fields gives an exact sequence

$$(6.2) \quad 0 \longrightarrow E_g \longrightarrow g^*T_X \longrightarrow Q_g \longrightarrow 0, \quad Q_g = \bigoplus_{i:m_i>0} g^*\mathcal{O}_X(D_i)|_{m_i[\infty]}.$$

To check exactness, choose simple normal crossings coordinates near $g(\infty)$, in which the components met by g are given by $x_i = 0$. In a source coordinate z centered at infinity,

$$x_i \circ g = z^{m_i} u_i(z), \quad u_i(0) \neq 0.$$

The corresponding logarithmic tangent vector maps to $z^{m_i} u_i(z) \partial/\partial x_i$. Thus the map is injective and its cokernel is the stated torsion sheaf. Away from infinity it is an isomorphism.

Ampleness gives $E_g \cong \bigoplus_{\nu=1}^n \mathcal{O}_{\mathbb{P}^1}(a_\nu)$ with $a_\nu \geq 1$, and therefore

$$(6.3) \quad H^1(\mathbb{P}^1, E_g) = H^1(\mathbb{P}^1, E_g(-2[0])) = 0.$$

Restriction to the divisor $2[0]$ means taking sections modulo t^2 in a local trivialization; it records their value and first derivative at 0. Since Q_g is a torsion sheaf, (6.2) also gives $H^1(\mathbb{P}^1, g^*T_X) = 0$. Deformation theory for maps with source $\mathbb{P}^1$ fixed therefore shows that H is smooth at g , with $T_g H = H^0(\mathbb{P}^1, g^*T_X)$; see [10, Proposition 1.6 and Theorem 1.9].

For $Z_i = m_i[\infty]$, the tangent space to the symmetric power has the intrinsic description

$$T_{Z_i} \text{Sym}^{m_i}(\mathbb{P}^1) \simeq H^0(Z_i, \mathcal{O}_{\mathbb{P}^1}(Z_i)|_{Z_i}) \simeq H^0(\mathbb{P}^1, g^*\mathcal{O}_X(D_i)|_{Z_i}),$$

where the second identification uses $g^*D_i = Z_i$; both sides are zero when $m_i = 0$. Under these identifications the differential of c is the quotient map

$$H^0(\mathbb{P}^1, g^*T_X) \longrightarrow H^0(\mathbb{P}^1, Q_g).$$

Locally it sends an infinitesimal variation of $x_i \circ g$ to its class modulo z^{m_i} , which is the infinitesimal displacement of the zero divisor of $x_i \circ g$. The vanishing $H^1(\mathbb{P}^1, E_g) = 0$ makes this differential surjective. Since the product of symmetric powers is smooth, c is smooth at g and its fiber M is smooth at g with the asserted tangent space.

First-jet evaluation. The chosen coordinate t identifies the space of first jets at 0 of germs of maps to X with TX . At the jet $J(g)$, the tangent space $T_{J(g)}TU$ is naturally identified with $H^0(\mathbb{P}^1, g^*T_X|_{2[0]})$. Since the divisor conditions are supported at ∞ , we have $E_g|_{2[0]} \simeq g^*T_X|_{2[0]}$. Under this identification and $T_g M \simeq H^0(\mathbb{P}^1, E_g)$, the differential dJ_g is the restriction map on global sections in the exact sequence

$$0 \longrightarrow E_g(-2[0]) \longrightarrow E_g \longrightarrow E_g|_{2[0]} \longrightarrow 0.$$

It is surjective by (6.3), so $J : M \rightarrow TU$ is smooth at g . Composing with the bundle projection $TU \rightarrow U$ gives the assertion about evaluation at 0. Restricting J to the fiber of evaluation at 0 over $g(0)$ gives the assertion about derivative evaluation. $\square$

Proof of Theorem 6.1. We construct the families in Definition 4.6, with target U , at every point of a nonempty Zariski open subset $U^+ \subset U$. Let $M^+ \subset M$ consist of the maps g for which E_g is ample; it contains f . On $\mathbb{P}^1$, ampleness is equivalent to $H^1(\mathbb{P}^1, E_g(-2[0])) = 0$. Let $e : M \times \mathbb{P}^1 \rightarrow X$ be the universal evaluation. The bundle

$$e^*T_X(-\log D) \otimes \text{pr}_{\mathbb{P}^1}^* \mathcal{O}_{\mathbb{P}^1}(-2[0])$$

is locally free on $M \times \mathbb{P}^1$ and flat over M . Since the projection $M \times \mathbb{P}^1 \rightarrow M$ is proper, semicontinuity of the fiberwise first cohomology shows that M^+ is Zariski open. By Lemma 6.2 the morphism

$$\mathrm{ev}_0 : M^+ \longrightarrow U$$

is smooth, so its image U^+ is a nonempty Zariski open subset of U .

By Lemma 6.2, M^+ is smooth. We now regard M^+ and U as complex manifolds. The evaluation family (6.1) restricts to a holomorphic map

$$\mathcal{F} : M^+ \times \mathbb{C} \longrightarrow U.$$

Fix $x \in U^+$ and $g \in M^+$ with $g(0) = x$. Derivative evaluation on $\mathrm{ev}_0^{-1}(x)$ is a submersion near g , so its image contains a nonempty Euclidean open subset of $T_x U$, and such a subset contains n linearly independent vectors. We may therefore choose maps $g_1, \dots, g_n \in M^+$ with $g_j(0) = x$ whose derivatives at 0 form a basis of $T_x U$.

For each j the holomorphic submersion ev_0 has a local holomorphic section $\sigma_j : V_j \rightarrow M^+$ with $\sigma_j(x) = g_j$. On the common neighborhood $V = \bigcap_j V_j$ of x set

$$h_j(y, t) = \mathcal{F}(\sigma_j(y), t), \quad (y, t) \in V \times \mathbb{C}.$$

By (6.1) these maps are holomorphic on $V \times \mathbb{C}$ with values in U , and $h_j(y, 0) = y$. Their derivatives $\partial_t h_j(y, 0)$ form a basis at x , hence also on a smaller common neighborhood of x , so the maps satisfy Definition 4.6.

Since X is smooth and connected, it is irreducible; its nonempty open subset $U = X \setminus D$ is therefore connected. The complement $U \setminus U^+$ is a proper closed algebraic, hence complex analytic, subset of U . Corollary 4.7 therefore implies that U is Oka. $\square$

The hypotheses on f in Theorem 6.1 allow $f(\mathbb{P}^1) \cap D = \emptyset$. Writing

$$f^* T_X(-\log D) \simeq \bigoplus_{\nu=1}^n \mathcal{O}_{\mathbb{P}^1}(a_\nu), \quad a_\nu \geq 1,$$

the determinant identity $\det T_X(-\log D) = \mathcal{O}_X(-(K_X + D))$ gives the necessary numerical condition

$$-(K_X + D) \cdot f_*[\mathbb{P}^1] = \deg f^* T_X(-\log D) \geq n.$$

Here $f_*[\mathbb{P}^1] = (\deg(f : \mathbb{P}^1 \rightarrow f(\mathbb{P}^1)))[f(\mathbb{P}^1)]$ is the pushforward one-cycle, and this numerical condition alone does not ensure that the pulled-back bundle is ample.

6.3. Geometric applications. Chen–Zhu’s algebraic notion of $\mathbb{A}^1$ -connectedness means that a family of nonconstant proper $\mathbb{A}^1$ -curves has Zariski-dense two-point evaluation [8, Definition 1.2]. This algebraic notion is distinct from motivic $\mathbb{A}^1$ -connectedness.

Corollary 6.3. *Let X be a connected complex projective manifold, and let $D \subset X$ be a reduced simple normal crossings divisor. If (X, D) is $\mathbb{A}^1$ -connected in the sense of Chen–Zhu, then $X \setminus D$ is Oka.*

Proof. Over $\mathbb{C}$, the differential of the two-point evaluation is surjective on a nonempty Zariski open subset of the smooth locus of its source, so the separability hypothesis in [8] holds. Chen–Zhu [8, Proposition 1.13(2)] therefore give a very free $\mathbb{A}^1$ -curve. For the divisorial log structure of (X, D) , their log tangent bundle is $T_X(-\log D)$, and such a curve has boundary contact only at the marked point ∞ [8, Definitions 1.11–1.12]. Its underlying map $f : \mathbb{P}^1 \rightarrow X$ satisfies the hypotheses of Theorem 6.1, which proves the corollary. $\square$

Ampleness of the boundary will also give holomorphic ellipticity in the applications below. If D is a nonzero ample effective Cartier divisor and $X \setminus D$ is Oka, choose $k > 0$ such that $\mathcal{O}_X(kD)$ is very ample. The linear system $|kD|$ embeds X in projective space, and the section with zero divisor kD cuts out a hyperplane. Then $X \setminus D$ is a closed subvariety of the affine

complement of that hyperplane, so it is affine and therefore Stein. Every Stein Oka manifold is holomorphically elliptic [36, 3.2.A, p. 879]; see also [31, Introduction].

When D has simple normal crossings, we call (X, D) *log Fano* if $-(K_X + D)$ is ample, following Maeda's convention [43, Definition 1.1]. Chen–Zhu's theorem gives $\mathbb{A}^1$ -connectedness when a smooth projective pair has a rationally connected fully free deepest stratum [8, Theorem 1.3]. In the applications below that stratum is $Z = D_1 \cap \cdots \cap D_r$. It is *fully free* if there are nonconstant free maps $a_1, \dots, a_s : \mathbb{P}^1 \rightarrow Z$ such that

$$c_{ij} = \deg a_j^*(\mathcal{O}_X(D_i)|_Z) \geq 0, \quad \text{rank}_{\mathbb{C}}((c_{ij})_{1 \leq i \leq r, 1 \leq j \leq s}) = r;$$

see [8, Remark 1.4]. Here a is free if $a^*T_Z \simeq \bigoplus_{\nu} \mathcal{O}_{\mathbb{P}^1}(b_{\nu})$ with $b_{\nu} \geq 0$.

Theorem 6.4. *Let X be a connected complex projective manifold of dimension at least two, and let $D \subset X$ be a smooth irreducible ample rationally connected divisor. Then $X \setminus D$ is Oka and holomorphically elliptic. The conclusion holds in particular for any smooth irreducible ample divisor $D \subset X$ such that $-(K_X + D)$ is ample.*

Proof. The smooth rationally connected divisor D carries a very free rational curve $a : \mathbb{P}^1 \rightarrow D$. When $\dim D = 1$ we have $D \simeq \mathbb{P}^1$, and the identity map under this identification suffices. Since D is ample,

$$\deg a^*\mathcal{O}_D(D) > 0.$$

Thus the 1×1 matrix $(\deg a^*\mathcal{O}_D(D))$ has rank one, and D is a fully free deepest stratum. Chen–Zhu [8, Theorem 1.3] give $\mathbb{A}^1$ -connectedness of (X, D) . Corollary 6.3 shows that $X \setminus D$ is Oka. Since D is a nonzero ample divisor, the affine-complement argument preceding the theorem also gives holomorphic ellipticity.

If $-(K_X + D)$ is ample, adjunction gives

$$-K_D = -(K_X + D)|_D.$$

Thus $-K_D$ is ample, so D is Fano and hence rationally connected [5, 40]. The first assertion applies. $\square$

Corollary 6.5. *Let $X \subset \mathbb{P}^N$ be a smooth complete intersection of multidegree $(d_1, \dots, d_c)$ and dimension at least two. If D is a smooth hypersurface section of degree $e \geq 1$ and*

$$d_1 + \cdots + d_c + e \leq N,$$

then $X \setminus D$ is Oka and holomorphically elliptic.

Proof. Write H for the hyperplane class of the ambient projective space. Then

$$-(K_X + D) = (N + 1 - d_1 - \cdots - d_c - e)H|_X.$$

The stated inequality makes this class ample. Since $\dim X \geq 2$, ampleness of D , Serre duality, and Kodaira vanishing give

$$H^1(X, \mathcal{O}_X(-D)) \simeq H^{\dim X - 1}(X, \mathcal{O}_X(K_X + D))^* = 0.$$

The sequence $0 \rightarrow \mathcal{O}_X(-D) \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_D \rightarrow 0$ then gives $H^0(D, \mathcal{O}_D) = \mathbb{C}$, so D is connected. Since D is smooth, it is irreducible, and Theorem 6.4 applies. $\square$

For several boundary components, we use their classes in $N^1(X)_{\mathbb{Q}}$, the space of Cartier divisor classes with rational coefficients modulo numerical equivalence.

Corollary 6.6. *Set $D = D_1 + \cdots + D_r$. Suppose (X, D) is a reduced simple normal crossings log Fano pair with $r \geq 1$, each D_i is ample, the classes $[D_1], \dots, [D_r]$ are linearly independent in $N^1(X)_{\mathbb{Q}}$, and $\dim X - r \geq 2$. Then $X \setminus D$ is Oka and holomorphically elliptic.*

Proof. Set $Z_0 = X$ and $Z_j = D_1 \cap \cdots \cap D_j$ for $1 \leq j \leq r$. Each Z_j is nonempty and smooth of dimension $\dim X - j$, and $D_j|_{Z_{j-1}}$ is ample. The connectedness argument in the proof of Corollary 6.5 therefore applies successively to these divisors and shows that $Z = Z_r$ is connected. Adjunction gives $-K_Z = -(K_X + D)|_Z$, so Z is Fano and rationally connected.

The divisors D_i are smooth and ample, their numerical classes are independent, and $\dim Z \geq 2$. Chen–Zhu [8, Theorem 1.5] therefore give $\mathbb{A}^1$ -connectedness of (X, D) . Corollary 6.3 shows that the complement is Oka. The sum $D = D_1 + \cdots + D_r$ is a nonzero ample divisor, so the same affine-complement argument gives holomorphic ellipticity. $\square$

For $X = \mathbb{P}^n$, the independence hypothesis in Corollary 6.6 forces $r = 1$, since $N^1(\mathbb{P}^n)_{\mathbb{Q}} = \mathbb{Q}[H]$ for the hyperplane class H .

For an example with two boundary components, take $X = \mathbb{P}^3 \times \mathbb{P}^3$ and general divisors $D_1 \in |\mathcal{O}_X(1, 1)|$ and $D_2 \in |\mathcal{O}_X(1, 2)|$. Both line bundles are very ample, so Bertini’s theorem gives smooth irreducible divisors meeting transversely. Their intersection is nonempty by ampleness and has dimension four. In the basis of the two hyperplane classes, the numerical classes $(1, 1)$ and $(1, 2)$ are independent, and $\mathcal{O}_X(-(K_X + D_1 + D_2)) \simeq \mathcal{O}_X(2, 1)$ is ample. Thus Corollary 6.6 shows that $X \setminus (D_1 \cup D_2)$ is Oka and holomorphically elliptic.

6.4. Examples and limitations. The following examples distinguish the curve criterion, algebraic $\mathbb{A}^1$ -connectedness, and ampleness of a log anticanonical $\mathbb{Q}$ -divisor.

A reduced simple normal crossings log Fano complement need not be $\mathbb{A}^1$ -connected. If $D \subset \mathbb{P}^2$ is the union of two distinct lines, then $\mathbb{P}^2 \setminus D \cong \mathbb{C} \times \mathbb{C}^*$ is homogeneous and Oka. It is not $\mathbb{A}^1$ -connected, since every algebraic map $\mathbb{A}^1 \rightarrow \mathbb{C}^*$ is constant. By contrast, the $\mathbb{C}^*$ factor has the holomorphic family $(y, t) \mapsto ye^t$.

For a smooth hypersurface $D \subset \mathbb{P}^n$ of degree $n + 1$,

$$\det T_{\mathbb{P}^n}(-\log D) = \mathcal{O}_{\mathbb{P}^n}(-(K_{\mathbb{P}^n} + D)) \cong \mathcal{O}_{\mathbb{P}^n}.$$

For every holomorphic map $f : \mathbb{P}^1 \rightarrow \mathbb{P}^n$, the bundle $f^*T_{\mathbb{P}^n}(-\log D)$ therefore has determinant of degree zero and cannot be ample, so Theorem 6.1 does not apply at this degree; Section 7 treats it by means of successive polar equations.

Nor does ampleness of $-(K_X + B)$ for an effective $\mathbb{Q}$ -divisor with coefficients in $(0, 1)$ suffice. On $\mathbb{P}^1$, the divisor $B = \frac{1}{2}([0] + [1] + [\infty])$ satisfies $\deg(-(K_{\mathbb{P}^1} + B)) = \frac{1}{2} > 0$, whereas $\mathbb{P}^1 \setminus \text{Supp } B$ is Kobayashi hyperbolic and hence not Oka.

7. SMOOTH HYPERSURFACE COMPLEMENTS

For a smooth hypersurface $D \subset \mathbb{P}^n$ of degree d , adjunction gives $K_D = \mathcal{O}_D(d - n - 1)$. Thus Theorem 6.4 gives the positive conclusion of Theorem 1.9 for $d \leq n$ when $n \geq 2$. For $3 \leq d \leq n + 1$, the construction below uses lines whose intersection multiplicity with D at a point of D is at least $d - 1$. It works for every smooth hypersurface in this degree range, and only the projection center is required to be general. Degrees one and two are treated directly. The obstruction in high degree is due to Carlson–Griffiths [7, Corollary (6.11)].

Lemma 7.1 (Polar transversality). *Let F be a homogeneous polynomial of degree d on $\mathbb{C}^N$, where $N \geq d \geq 3$, and suppose that $X = \{F = 0\} \subset \mathbb{P}^{N-1}$ is smooth. There is a nonempty Zariski open subset $\Omega \subset \mathbb{P}^{N-1} \setminus X$ such that, for every $[a] \in \Omega$ and every nonzero representative $a = (a_1, \dots, a_N)$, the equations*

$$Z_a = \{F = D_a F = \cdots = D_a^{d-2} F = 0\} \subset \mathbb{P}^{N-1}, \quad D_a = \sum_{i=1}^N a_i \frac{\partial}{\partial v_i}$$

have linearly independent differentials along Z_a in local projective coordinates. In particular, every nonempty Z_a is a complex submanifold of dimension $N - d$. Moreover, the incidence set

$$\mathcal{I} = \{([a], [v]) \in \Omega \times X : D_a F(v) = \cdots = D_a^{d-2} F(v) = 0\}$$

is a complex submanifold of $\Omega \times X$, and the projection $\pi : \mathcal{I} \rightarrow \Omega$ is a holomorphic submersion. Every nonempty fiber has dimension $N - d$.

Proof. Generic projection. Fix a hyperplane $L \simeq \mathbb{P}^{N-2}$. Projection $p_a : X \rightarrow L$ from a point $[a] \notin X \cup L$ is finite: a positive-dimensional fiber would contain the entire line through $[a]$, contradicting $[a] \notin X$. The kernel of its differential has dimension at most one, since the ambient projection has one-dimensional fibers. Mather's generic-projection theorem, in the formulation of Alzati–Ballico–Ottaviani [2, Theorem 1.1 and Section 2, step (2)], applies with ambient projective dimension $N - 1$, codimension one, and center dimension zero. For $1 \leq k \leq d - 2 \leq N - 2$, let $I_k = (1, \dots, 1)$ have k entries, and let $J^k(X, L)$ be the bundle of k -jets of holomorphic maps from X to L . The theorem gives a nonempty Zariski open set of centers for which the k -jet map $j^k p_a : X \rightarrow J^k(X, L)$ is transverse to the Thom–Boardman stratum Σ^{I_k} , of codimension k in this equidimensional corank-one case. The repeated symbol records a one-dimensional kernel at each successive critical locus. The space of centers $\mathbb{P}^{N-1} \setminus (X \cup L)$ is irreducible, so intersecting these finitely many nonempty open sets gives a nonempty Zariski open set $\Omega \subset \mathbb{P}^{N-1} \setminus (X \cup L)$ on which all these transversality conditions hold. The hypersurface X remains fixed.

Identification with the polar equations. Near a common zero $[v]$, choose a linear form ℓ with $\ell(a) = 0$ and $\ell(v) \neq 0$. On the affine chart $\ell = 1$, take coordinates (z, u, t) , with $z \in \mathbb{C}^{N-3}$, so that translation by a is translation in t and the projection forgets t . The equation of X is $P(z, u, t) = 0$, where $\partial_t^j P = (D_a^j F)|_{\ell=1}$. Since $\partial_t P = 0$ at the point and X is smooth, we can choose u so that $\partial_u P \neq 0$ there and write $u = g(z, t)$. After shrinking the coordinate neighborhood,

$$A(z, t) = \partial_u P(z, g(z, t), t)$$

is nowhere zero. Set $q_j = (\partial_t^j P)(z, g(z, t), t)$ and $h_j = \partial_t^j g$. The derivatives of P are taken with u fixed before restriction to the graph. Repeated differentiation of $P(z, g(z, t), t) = 0$ gives

$$q_j + Ah_j \in (h_1, \dots, h_{j-1}), \quad (q_1, \dots, q_j) = (h_1, \dots, h_j).$$

These relations hold in the local holomorphic ring at the point for $1 \leq j \leq d - 2$. At a common zero, the differentials of the q_j and h_j are related by an invertible triangular matrix. Write $S_k = \{h_1 = \dots = h_k = 0\}$ in the graph coordinates, where the projection has the adapted form $p_a(z, t) = (z, g(z, t))$. In the corresponding adapted jet coordinates, the local equations for Σ^{I_k} are the vanishing of the first k successive derivatives in the kernel direction. We identify their pullbacks inductively. For $k = 1$, the critical locus is $S_1 = \{h_1 = 0\}$; jet transversality gives $dh_1 \neq 0$ there, so S_1 is smooth and the kernel of dp_a is $\mathbb{C} \partial/\partial t$ on S_1 .

Suppose that $1 \leq k \leq d - 3$, that S_k is the pullback of Σ^{I_k} , and that $d(h_1, \dots, h_k)$ has rank k on S_k . For $p \in S_k$, the identities $\partial_t h_i = h_{i+1}$ give

$$\frac{\partial}{\partial t} \in T_p S_k \iff h_{k+1}(p) = 0.$$

Thus the next corank-one locus is S_{k+1} . Its local defining functions in the adapted jet coordinates pull back to $h_1, \dots, h_{k+1}$, so transversality to $\Sigma^{I_{k+1}}$ gives rank $d(h_1, \dots, h_{k+1}) = k + 1$ on S_{k+1} . This proves by induction that rank $d(h_1, \dots, h_k) = k$ on S_k for every $1 \leq k \leq d - 2$, including $d = 3$, when only the initial step is needed. The triangular change of generators gives the same rank for $(q_1, \dots, q_k)$. Hence the polar equations have linearly independent differentials along their common zero set in X . By the holomorphic implicit function theorem, they cut out a complex submanifold of codimension $d - 2$ in X .

Each equation $D_a^j F(v) = 0$ is bihomogeneous of bidegree $(j, d - j)$ in (a, v) , so the incidence set is independent of the chosen projective representatives. The differentials in the v -variables are surjective along every fiber by the preceding rank calculation. The holomorphic implicit function theorem therefore makes $\mathcal{I}$ a complex submanifold and π a holomorphic submersion; every nonempty fiber has dimension $N - d$. $\square$

Lemma 7.2 (Polar complete intersections). *With Ω , Z_a , $\mathcal{I}$, and π as in Lemma 7.1, every Z_a is nonempty and is a projective complete intersection of multidegree $(d, d-1, \dots, 2)$ and degree $d!$. No hyperplane in $\mathbb{P}^{N-1}$ contains Z_a . Consequently, $\pi : \mathcal{I} \rightarrow \Omega$ is surjective.*

Proof. Let J be the ideal generated by $F, D_a F, \dots, D_a^{d-2} F$ in $S = \mathbb{C}[v_1, \dots, v_N]$. Its projective zero set is nonempty: otherwise $\sqrt{J}$ would be the irrelevant ideal of height N , contrary to the height theorem for an ideal generated by $d-1 < N$ elements. The dimension calculation above shows that $\text{ht } J = d-1$. Thus the generators form a regular sequence, and S/J is Cohen–Macaulay of dimension $N-d+1 \geq 1$. Write $\mathfrak{m} = (v_1, \dots, v_N)$. The saturation J^{sat} consists of the polynomials $b \in S$ such that $\mathfrak{m}^r b \subset J$ for some r . A general linear form ρ is a nonzerodivisor on S/J . If $b \in J^{\text{sat}}$, then $\rho^r b \in J$ for some r , whence $b \in J$; therefore $J = J^{\text{sat}}$. The projective scheme cut out by these equations is smooth, hence reduced. Its saturated homogeneous ideal is consequently the ideal of homogeneous polynomials vanishing on its points; saturation rules out an additional component supported only at the cone vertex. In particular, a linear form vanishing at all its points belongs to J . But J has no element of degree one, because all its generators have degree at least two. Thus no hyperplane contains its points. Its degree as a complete intersection is $d(d-1) \cdots 2 = d!$. The nonemptiness of every Z_a makes π surjective. $\square$

When $N = d$, the incidence projection π is a finite unramified holomorphic covering of degree $d!$. Indeed, it is proper because $\mathcal{I}$ is closed in $\Omega \times X$ and X is compact. Its fibers are zero-dimensional, so the holomorphic submersion π is a local biholomorphism. Properness then makes it a finite covering. Each fiber consists of $d!$ distinct points.

Proof of Theorem 1.9. The range $3 \leq d \leq n+1$.

Write $D = \{F = 0\}$ for a homogeneous polynomial F of degree d on $\mathbb{C}^{n+1}$. Suppose that $3 \leq d \leq n+1$. Apply Lemmas 7.1 and 7.2 with $N = n+1$ and $X = D$, and write $y_0 = [a_0] \in \Omega$ with $a_0 \neq 0$. Since Z_{a_0} spans $\mathbb{P}^n$, we can choose points $[v_1^0], \dots, [v_n^0] \in Z_{a_0}$ whose representatives have classes forming a basis of $\mathbb{C}^{n+1}/\mathbb{C}a_0$. The holomorphic submersion π has local holomorphic sections through the points $([a_0], [v_j^0]) \in \mathcal{I}$. Taking nonzero holomorphic lifts of their second components, together with a local holomorphic lift of the inclusion into $\mathbb{P}^n$, gives maps

$$a, v_1, \dots, v_n : V \longrightarrow \mathbb{C}^{n+1} \setminus \{0\}, \quad [a(y)] = y, \quad [v_j(y)] \in Z_{a(y)},$$

on a common coordinate neighborhood $V \subset \Omega$ of y_0 . After shrinking V , the classes of $v_1(y), \dots, v_n(y)$ modulo $\mathbb{C}a(y)$ form a basis for every $y \in V$.

For each fixed y , write $a = a(y)$ and $v = v_j(y)$. For $s \neq 0$, homogeneity and Taylor expansion give

$$F(a + sv) = s^d F(v + a/s) = \sum_{\ell=0}^d \frac{s^{d-\ell}}{\ell!} D_a^\ell F(v).$$

The resulting polynomial identity also holds at $s = 0$. In particular,

$$[s^k]F(a + sv) = \frac{D_a^{d-k} F(v)}{(d-k)!} = 0 \quad (2 \leq k \leq d),$$

where $[s^k]$ denotes the coefficient of s^k . Consequently,

$$F(a + sv) = F(a) + s dF_a(v), \quad F(ra + sv) = r^{d-1}(rF(a) + s dF_a(v)).$$

Let $L_{a,v} = \{[ra + sv] : [r : s] \in \mathbb{P}^1\}$ and write $\lambda = dF_a(v)$. Since $F(a) \neq 0$, this line is not contained in D , and the displayed factorization gives its intersection divisor:

$$D|_{L_{a,v}} = \begin{cases} (d-1)[v] + [\lambda a - F(a)v], & \lambda \neq 0, \\ d[v], & \lambda = 0. \end{cases}$$

In particular, the line meets D in at most two distinct points.

The function

$$E(\lambda, t) = \begin{cases} \frac{e^{\lambda t} - 1}{\lambda}, & \lambda \neq 0, \\ t, & \lambda = 0, \end{cases} \quad E(\lambda, t) = \sum_{k=0}^{\infty} \frac{\lambda^k t^{k+1}}{(k+1)!},$$

is entire on $\mathbb{C}^2$, since the series converges locally uniformly, and satisfies $1 + \lambda E(\lambda, t) = e^{\lambda t}$. Its images are

$$E(0, \mathbb{C}) = \mathbb{C}, \quad E(\lambda, \mathbb{C}) = \mathbb{C} \setminus \{-1/\lambda\} \quad (\lambda \neq 0).$$

When $\lambda = 0$, the line complement is $\mathbb{C}$, parametrized by $s = F(a)t$; when $\lambda \neq 0$, it is $\mathbb{C}^*$, and $s = F(a)E(\lambda, t)$ omits exactly the finite value $-F(a)/\lambda$. These formulas give entire maps onto the line complements and remain jointly holomorphic in λ and t at $\lambda = 0$. Write $c(y) = F(a(y))$ and, for $j = 1, \dots, n$, write $\lambda_j(y) = dF_{a(y)}(v_j(y))$. The maps

$$h_j(y, t) = [a(y) + c(y)E(\lambda_j(y), t)v_j(y)], \quad (y, t) \in V \times \mathbb{C},$$

satisfy

$$F(a(y) + c(y)E(\lambda_j(y), t)v_j(y)) = c(y)e^{\lambda_j(y)t} \neq 0.$$

Hence $h_j : V \times \mathbb{C} \rightarrow \mathbb{P}^n \setminus D$ is well defined and holomorphic, with $h_j(y, 0) = y$. Let $q : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{P}^n$ be projectivization. Then

$$\partial_t h_j(y, 0) = dq_{a(y)}(c(y)v_j(y)).$$

Since $\ker dq_{a(y)} = \mathbb{C}a(y)$ and $c(y) \neq 0$, these velocities form a basis of $T_y \mathbb{P}^n$. In this construction $a(y)$ and $v_j(y)$ are held fixed as t varies.

The complement $\mathbb{P}^n \setminus D$ is connected: the line joining two distinct points of the complement is not contained in D , hence meets D in a finite set, and the complement of that set in the line is connected. Write $Y = \mathbb{P}^n \setminus D$ and $A = Y \setminus \Omega$. Then A is a proper closed algebraic, hence complex analytic, subset of the connected manifold Y . The construction works at every point of $Y \setminus A$. The general case of Corollary 4.7 therefore shows that Y is Oka.

Degrees one and two. For $d = 2$, the smooth quadric $\{F = 0\}$ spans $\mathbb{P}^n$; when $n = 1$, it consists of two distinct points, which span $\mathbb{P}^1$. At any $y_0 \notin D$, choose a local holomorphic lift $a(y)$ of y and nonzero constant vectors $v_1, \dots, v_n$ with $F(v_j) = 0$ whose classes form a basis of $\mathbb{C}^{n+1}/\mathbb{C}a(y_0)$. After shrinking the neighborhood V of y_0 , use the same formula for h_j . The identity $F(a + sv) = F(a) + s dF_a(v)$ and the preceding argument apply unchanged. For $d = 1$, the complement is $\mathbb{C}^n$. These cases include $\mathbb{P}^1$ minus two points, which is $\mathbb{C}^*$.

The range $d \geq n + 2$. If $d \geq n + 2$, then $\mathcal{O}_{\mathbb{P}^n}(K_{\mathbb{P}^n} + D) \cong \mathcal{O}_{\mathbb{P}^n}(d - n - 1)$ is ample. The smooth divisor $D \in |\mathcal{O}_{\mathbb{P}^n}(d)|$ therefore satisfies the hypotheses of Carlson–Griffiths [7, Section 6(a), Corollary (6.11)], with $V = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(d)$. Their corollary implies that every entire map $\mathbb{C}^n \rightarrow \mathbb{P}^n \setminus D$ has rank less than n everywhere.

By Theorem 1.3, an Oka n -fold is strongly dominable and hence admits an entire map from $\mathbb{C}^n$ with differential of rank n at some point. This contradicts the rank bound, so $\mathbb{P}^n \setminus D$ is not Oka in this degree range.

Since D is an ample effective divisor, the affine-complement argument in Section 6 shows that $\mathbb{P}^n \setminus D$ is holomorphically elliptic whenever it is Oka. Together with the implication from holomorphic ellipticity to the Oka property, this proves that the three stated conditions are equivalent. $\square$

For a smooth quartic surface in $\mathbb{P}^3$ and a smooth quintic threefold in $\mathbb{P}^4$, Lemma 7.2 gives respectively 24 and 120 distinct points spanning the ambient projective space for every center in a nonempty Zariski open set.

The smoothness assumption in the degree criterion is essential. For $m \geq 3$, the reduced divisor

$$D_m = \{Z(YZ^{m-1} - X^m) = 0\} \subset \mathbb{P}^2$$

is the union of the line at infinity and the closure of $y = x^m$, and has degree $m + 1 > 3$. The graph closure is singular at $[0 : 1 : 0]$: in the chart $Y = 1$, with $x = X/Y$ and $z = Z/Y$, its equation is $z^{m-1} - x^m = 0$, whose first derivatives vanish at $(0, 0)$ for $m \geq 3$. In the chart $Z = 1$, the polynomial automorphism $(x, y) \mapsto (x, y - x^m)$ identifies $\mathbb{P}^2 \setminus D_m$ with $\mathbb{C} \times \mathbb{C}^*$, so this complement is Oka.

8. ABELIAN FIBRATIONS AND CALABI–YAU THREEFOLDS

Consider a projective fibration over $\mathbb{P}^1$ with a section and abelian fibers over a nonempty Zariski open set. Complete vertical flows move the section through every such fiber lying over an affine base chart. Their joint action produces a single entire map that covers these fibers and is locally biholomorphic at every point mapping to one of them. Corollary 4.5 then gives the Oka property. The construction is not restricted to Calabi–Yau manifolds.

8.1. The abelian-fibration theorem.

Theorem 8.1. *Let X be a smooth connected complex projective manifold of dimension $r + 1$, where $r \geq 1$, and let $f : X \rightarrow \mathbb{P}^1$ be a surjective morphism with an algebraic section $\sigma : \mathbb{P}^1 \rightarrow X$. Suppose that f is smooth over a nonempty Zariski open set $B^\circ \subset \mathbb{P}^1$ and that each fiber $X_b = f^{-1}(b)$, $b \in B^\circ$, is a connected abelian variety of dimension r . For every $b_\infty \in \mathbb{P}^1$ and every algebraic isomorphism $\alpha : \mathbb{C} \rightarrow \mathbb{P}^1 \setminus \{b_\infty\}$, there is a holomorphic map*

$$F : \mathbb{C} \times \mathbb{C}^r \longrightarrow X$$

such that

$$f(F(z, t)) = \alpha(z), \quad U := f^{-1}(B^\circ \setminus \{b_\infty\}) \subset F(\mathbb{C}^{r+1}),$$

and dF_q is an isomorphism for every $q \in F^{-1}(U)$. Consequently, X is Oka.

Proof. We first globalize vertical vector fields over an affine base chart and then integrate them. The fields are defined on the full inverse image of the chart, including the fibers outside B° . All sheaves below are algebraic.

Global vertical vector fields. Consider

$$\mathcal{V} = \ker(df : T_X \longrightarrow f^*T_{\mathbb{P}^1}), \quad \mathcal{E} = f_*\mathcal{V}.$$

The kernel $\mathcal{V}$ is coherent, and proper direct image makes $\mathcal{E}$ coherent on $\mathbb{P}^1$ [48, Tag 02O5]. It is also torsion-free. Indeed, let h be a nonzero local regular function on the base. Multiplication by h on $\mathcal{E}$ corresponds to multiplication by f^*h on a vertical field. The smooth connected manifold X is integral, and surjectivity of f makes f^*h nonzero; hence no nonzero local section of T_X is annihilated by f^*h . Finally, a coherent torsion-free sheaf on the smooth curve $\mathbb{P}^1$ is locally free, since finite torsion-free modules over a discrete valuation ring are free.

Write $X^\circ = f^{-1}(B^\circ)$. On X° the sheaf $\mathcal{V}$ is the relative tangent bundle T_{X°/B° . Every holomorphic vector field on an abelian variety A is translation-invariant: in the translation trivialization of T_A , its coefficients are holomorphic functions on the compact connected manifold A , hence constant. Thus

$$H^0(A, T_A) \longrightarrow T_x A \quad \text{is an isomorphism for every } x \in A,$$

$$h^q(A, T_A) = r \binom{r}{q} \quad (0 \leq q \leq r).$$

These dimensions are constant on B° , and the relative tangent bundle is flat over B° because f is smooth there. By cohomology and base change, the derived direct image of T_{X°/B° is represented locally on B° by a finite locally free complex compatible with passage to every fiber [48, Tags 07VK and 07VL]. Constancy of all the fiber cohomology dimensions makes the

differentials of this complex have locally constant rank. Its degree-zero cohomology therefore commutes with passage to fibers, and

$$\mathcal{E} \otimes k(b) \simeq H^0(X_b, T_{X_b}) \quad (b \in B^\circ), \quad \text{rank } \mathcal{E} = r.$$

On each fiber over B° , the evaluation morphism is the isomorphism displayed above. It is therefore an isomorphism of vector bundles

$$(8.1) \quad f^*(\mathcal{E}|_{B^\circ}) \longrightarrow T_{X^\circ/B^\circ}.$$

We use this isomorphism only over B° ; over the remaining fibers we need the evaluated fields merely to be defined.

Let $B_0 = \mathbb{P}^1 \setminus \{b_\infty\}$ and $X_0 = f^{-1}(B_0)$. The coordinate α identifies B_0 with the affine line. Hence the finite module $\Gamma(B_0, \mathcal{E})$ is torsion-free over the principal ideal domain $\mathbb{C}[z]$, and is therefore free of rank r . A basis $e_1, \dots, e_r$ gives, by evaluation, algebraic vector fields $V_1, \dots, V_r$ on all of X_0 satisfying $df(V_j) = 0$. These are holomorphic fields on the full inverse image of the affine chart. By (8.1), their values form a basis of $T_x X_b$ for every $b \in B_0 \cap B^\circ$ and $x \in X_b$. On each such fiber they are translation-invariant and commute. The brackets $[V_i, V_j]$ therefore vanish on a dense open subset, and hence on all of X_0 .

Complete flows and dominability. Each V_j has a complete jointly holomorphic flow $\Phi_j : \mathbb{C} \times X_0 \rightarrow X_0$. Local integral curves preserve f and therefore remain in one compact fiber. More uniformly, if $K \subset B_0$ is a compact neighborhood of a base point, then $f^{-1}(K)$ is compact. The local holomorphic ODE theorem and a finite coordinate cover give a common small complex-time disc for all initial points in this set, with holomorphic dependence on the initial point. Subdividing the straight segment from 0 to any prescribed time gives successive continuation, because every intermediate point remains in the same compact fiber. Uniqueness makes these continuations compatible and yields both the flow law and joint holomorphic dependence. This continuation argument takes place on the smooth total space X_0 and applies equally over singular fibers.

The vanishing brackets and uniqueness imply that the complete flows commute. Their composition is a holomorphic $\mathbb{C}^r$ -action

$$\Psi(t, x) = (\Phi_r(t_r, \cdot) \circ \dots \circ \Phi_1(t_1, \cdot))(x)$$

on X_0 , preserving f and satisfying $\partial_{t_j} \Psi(t, x) = V_j(\Psi(t, x))$. Every orbit in a fiber over $B_0 \cap B^\circ$ is open, because its orbit map has surjective differential. Since the fiber is connected and the orbits partition it into disjoint open sets, the action is transitive.

The section now gives the entire map

$$F(z, t) = \Psi(t, \sigma(\alpha(z))).$$

It is holomorphic on all of $\mathbb{C}^{r+1}$ and satisfies $f \circ F = \alpha \circ \text{pr}_z$. Transitivity proves $U \subset F(\mathbb{C}^{r+1})$. For $q = (z, t) \in F^{-1}(U)$, the vectors

$$\partial_{t_j} F(q) = V_j(F(q)), \quad 1 \leq j \leq r,$$

form a basis of the vertical tangent space, while

$$df_{F(q)}(\partial_z F(q)) = d\alpha_z(\partial/\partial z) \neq 0.$$

Hence dF_q is an isomorphism.

Finally,

$$A = X \setminus U = f^{-1}((\mathbb{P}^1 \setminus B^\circ) \cup \{b_\infty\})$$

is a proper closed algebraic subset. For every $x \in U$, a preimage $a \in \mathbb{C}^{r+1}$ gives an entire map $w \mapsto F(a + w)$ taking 0 to x with invertible differential there. Thus X is dominable at every point of $X \setminus A$, and Corollary 4.5 implies that X is Oka. $\square$

Taking $r = 2$ in Theorem 8.1 gives the following Calabi–Yau threefold case. Let X be a smooth connected projective Calabi–Yau threefold, and let $f : X \rightarrow \mathbb{P}^1$ be a surjective morphism with an algebraic section. If f is smooth over a nonempty Zariski open set B° and has connected abelian surfaces as fibers there, then X is Oka. Moreover, for every $b_\infty \in \mathbb{P}^1$, there is an entire map $F : \mathbb{C}^3 \rightarrow X$ whose image contains

$$U = f^{-1}(B^\circ \setminus \{b_\infty\})$$

and whose differential is invertible on $F^{-1}(U)$.

8.2. Schoen fiber products. Schoen’s fiber products [47] furnish concrete Calabi–Yau examples. Here a *Schoen threefold* means a fiber product satisfying the hypotheses below. The geometric part of the proof establishes the Calabi–Yau properties; the fibration theorem then supplies the entire map and the Oka property.

Theorem 8.2. *For $i = 1, 2$, let $p_i : S_i \rightarrow \mathbb{P}^1$ be relatively minimal elliptic fibrations with connected fibers and sections on smooth projective rational surfaces, and denote their sets of singular values by $\Delta_i \subset \mathbb{P}^1$. Assume that $\Delta_1 \cap \Delta_2 = \emptyset$. Then the fiber product*

$$X = S_1 \times_{\mathbb{P}^1} S_2$$

is a smooth simply connected projective threefold with $K_X \simeq \mathcal{O}_X$ and $H^1(X, \mathcal{O}_X) = H^2(X, \mathcal{O}_X) = 0$. It is Oka. For the induced morphism $p : X \rightarrow \mathbb{P}^1$ and every $b_\infty \in \mathbb{P}^1$, there is a holomorphic map $F : \mathbb{C}^3 \rightarrow X$ whose image contains

$$U = p^{-1}(\mathbb{P}^1 \setminus (\Delta_1 \cup \Delta_2 \cup \{b_\infty\}))$$

and whose differential is invertible at every point of $F^{-1}(U)$.

Proof of Theorem 8.2. Smoothness and Calabi–Yau geometry. We regard X as the subset of $S_1 \times S_2$ defined by $p_1(x_1) = p_2(x_2)$. Write $s_i : \mathbb{P}^1 \rightarrow S_i$ for the sections and $p : X \rightarrow \mathbb{P}^1$ for the induced morphism. Write $M = S_1 \times S_2$ and denote the diagonal in $\mathbb{P}^1 \times \mathbb{P}^1$ by Δ . At every point of X , at least one of dp_1 and dp_2 is surjective. Hence $p_1 \times p_2 : M \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ is transverse to Δ , and $X = (p_1 \times p_2)^{-1}(\Delta)$ is a smooth threefold. It is projective as a closed submanifold of M , and it is connected because both the base and the fibers of p are connected.

A section meets every fiber with intersection number one. If a fiber were mF' with an integer $m \geq 2$ and an effective divisor F' , then

$$1 = s_i(\mathbb{P}^1) \cdot (mF') = m(s_i(\mathbb{P}^1) \cdot F'),$$

contrary to the integrality of $s_i(\mathbb{P}^1) \cdot F'$. Thus there are no multiple fibers. The canonical bundle formula for relatively minimal elliptic surfaces gives $K_{S_i} \simeq p_i^*(K_{\mathbb{P}^1} \otimes \mathcal{L}_i)$, where $\mathcal{L}_i = (R^1(p_i)_* \mathcal{O}_{S_i})^\vee$ has degree $\chi(\mathcal{O}_{S_i}) = 1$. Thus $K_{S_i} \simeq p_i^* \mathcal{O}_{\mathbb{P}^1}(-1)$; see also [12, Section 3.1, p. 506]. The diagonal has bidegree $(1, 1)$, so $\mathcal{O}_M(X) \simeq K_M^{-1}$. Adjunction gives $K_X \simeq \mathcal{O}_X$. Rationality and the Künneth formula give $H^j(M, \mathcal{O}_M) = 0$ for $j > 0$, and Serre duality gives $H^j(M, K_M) = 0$ for $j < 4$. The exact sequence $0 \rightarrow K_M \rightarrow \mathcal{O}_M \rightarrow \mathcal{O}_X \rightarrow 0$ therefore gives $H^1(X, \mathcal{O}_X) = H^2(X, \mathcal{O}_X) = 0$.

Simple connectedness. To prove that X is simply connected, we follow Gachet–Lin–Wang [34, Lemma 4.8]. Write $B^\circ = \mathbb{P}^1 \setminus (\Delta_1 \cup \Delta_2)$, and write $S_{i,b} = p_i^{-1}(b)$. Fix $b \in B^\circ$ and use $(s_1(b), s_2(b))$ as the base point. The smooth proper fibration $p^{-1}(B^\circ) \rightarrow B^\circ$ has the section (s_1, s_2) and fiber $X_b = S_{1,b} \times S_{2,b}$. Its fundamental group is generated by the images of $\pi_1(X_b)$ and of $\pi_1(B^\circ)$ under the fiber inclusion and the section, respectively. Since loops can be perturbed off the omitted complex divisor, the inclusion $p^{-1}(B^\circ) \hookrightarrow X$ induces a surjection on fundamental groups. The section extends over $\mathbb{P}^1$, so the image of $\pi_1(B^\circ)$ in $\pi_1(X)$ is trivial. The two factors of $\pi_1(X_b)$ also have trivial image: their inclusions into X factor through the simply connected rational surfaces by

$$x_1 \mapsto (x_1, s_2(p_1(x_1))), \quad x_2 \mapsto (s_1(p_2(x_2)), x_2).$$

Hence $\pi_1(X) = 1$.

The Oka property and the entire map. The morphism $p : X \rightarrow \mathbb{P}^1$ has the algebraic section $b \mapsto (s_1(b), s_2(b))$. Over $B^\circ = \mathbb{P}^1 \setminus (\Delta_1 \cup \Delta_2)$ it is smooth, with connected fibers $S_{1,b} \times S_{2,b}$, which are abelian surfaces. Theorem 8.1, with $r = 2$, proves both the Oka property and the stated assertions about F . $\square$

Such pairs of elliptic surfaces exist. Blowing up the nine distinct base points of a general pencil of plane cubics gives a relatively minimal rational elliptic surface, and every exceptional curve is a section [12, Section 2, p. 503]. Starting from two such surfaces, choose $\gamma \in \mathrm{PGL}_2(\mathbb{C})$ with $\gamma(\Delta_2) \cap \Delta_1 = \emptyset$; a general γ works because both sets are finite. Replacing p_2 by $\gamma \circ p_2$ and a section s_2 by $s_2 \circ \gamma^{-1}$ gives the required pair.

These threefolds are not rationally connected. Indeed, if X were rationally connected, it would admit a very free holomorphic map $u : \mathbb{P}^1 \rightarrow X$, for which

$$\deg u^* K_X^{-1} = \deg u^* TX \geq 3.$$

This contradicts $K_X \simeq \mathcal{O}_X$.

The projective Oka manifolds obtained from rational curves and from abelian fibrations also admit dominating holomorphic sprays.

Corollary 8.3. *Every smooth complex projective rationally connected manifold is holomorphically elliptic. The same holds for every manifold satisfying the hypotheses of Theorem 8.1, and hence for every Schoen threefold in Theorem 8.2.*

Proof. Theorems 1.8 and 8.1 give the Oka property. Since the manifolds are projective, Forstnerič–Lárusson’s theorem [31, Theorem 1.1] gives holomorphic ellipticity. $\square$

9. SPACES OF RATIONAL MAPS

For $d \geq 1$, let

$$R_d = \mathrm{Hol}_d(\mathbb{P}^1, \mathbb{P}^1)$$

be the $(2d+1)$ -dimensional manifold of rational maps of degree d . The space R_2 is homogeneous and hence Oka [38, Theorem 1.1 and Corollary 1.2]. Hanysz proved that R_3 is strongly dominable and strongly $\mathbb{C}$ -connected [38, Theorems 1.7 and 1.8]. Xie subsequently proved that R_3 is holomorphically elliptic, hence Oka [51, Theorem 1.3]. For arbitrary degree, the Euclidean algorithm produces an Oka open set in the normalized evaluation fiber. The union theorem gives the Oka property of the whole fiber, and the evaluation fiber bundle then shows that R_d is Oka.

For $f \in R_d$ satisfying $f(\infty) = 0$, the coprime representation $f = p/q$ has a unique normalization in which q is monic of degree d and $\deg p < d$. Thus, via $(q, p) \mapsto p/q$, the fiber of evaluation at infinity over 0 is identified biholomorphically with

$$\mathcal{R}_d = \{(q, p) : q \text{ monic of degree } d, \deg p < d, \mathrm{Res}(q, p) \neq 0\},$$

a smooth principal open subset of $\mathbb{C}^{2d}$.

Proof of Proposition 1.10. The Euclidean-algorithm cell. The Euclidean algorithm supplies the coordinates of the Oka open set: linear partial quotients contribute parameters in $\mathbb{C}$, and normalization factors contribute parameters in $\mathbb{C}^*$. This is a monic-remainder version of the linear-partial-quotient case of Cazanave’s continued-fraction decomposition [18, Lemma A.1]. The cited appendix represents pointed rational functions as A/B ; in our notation this is the reciprocal q/p . We give the monic-remainder coordinates and their inverses explicitly below.

For $b_1, \dots, b_d \in \mathbb{C}$ and $a, c_1, \dots, c_{d-1} \in \mathbb{C}^*$, set

$$(9.1) \quad r_d = 1, \quad r_{d-1} = z - b_d, \quad r_{i-1} = (z - b_i)r_i + c_i r_{i+1} \quad (i = d-1, \dots, 1).$$

Then r_i is monic of degree $d - i$ for $0 \leq i \leq d$, and

$$\gcd(r_{i-1}, r_i) = \gcd(r_i, r_{i+1}) = 1 \quad (1 \leq i \leq d-1),$$

by induction from $r_d = 1$. Consequently $(q, p) = (r_0, ar_1)$ belongs to $\mathcal{R}_d$.

Let $U_d \subset \mathcal{R}_d$ consist of the pairs for which $\deg p = d - 1$ and the successive nonzero monic remainders have degrees $d - 2, \dots, 0$. Monic division recovers the parameters in (9.1) uniquely:

$$a = \text{lc}(p), \quad r_0 = q, \quad r_1 = p/a,$$

and, for $i = 1, \dots, d - 1$,

$$r_{i-1} = (z - b_i)r_i + \tilde{r}_{i+1}, \quad c_i = \text{lc}(\tilde{r}_{i+1}), \quad r_{i+1} = \tilde{r}_{i+1}/c_i.$$

Here lc denotes the leading coefficient, and the terminal relation $r_{d-1} = z - b_d$ recovers b_d . At each division the coefficients are regular functions on the open set obtained at the preceding step, and the next degree condition is the nonvanishing of the coefficient of z^{d-i-1} in $\tilde{r}_{i+1}$. Hence U_d is Zariski open, and (9.1) and monic division give algebraic inverses establishing

$$(9.2) \quad U_d \cong \mathbb{C}^d \times (\mathbb{C}^*)^d.$$

For $d = 1$, the recursion is empty: $r_1 = 1$, $r_0 = z - b_1$, and $p = a \neq 0$, so $U_1 = \mathcal{R}_1 \cong \mathbb{C} \times \mathbb{C}^*$. In particular, U_d is nonempty and Oka, since complex Lie groups are Oka [25, Proposition 5.5.1]. The principal open set $\mathcal{R}_d \subset \mathbb{C}^{2d}$ is irreducible and connected, and (9.2) exhibits U_d as a dense Zariski open Oka subset. Theorem 1.7 therefore gives that $\mathcal{R}_d$ is Oka.

The evaluation bundle. Evaluation at infinity gives a holomorphic fiber bundle

$$(9.3) \quad \text{ev}_\infty : R_d \longrightarrow \mathbb{P}^1$$

with fiber $\mathcal{R}_d$. Indeed, locally in $b \in \mathbb{P}^1$, choose Möbius transformations g_b depending holomorphically on b and satisfying $g_b(0) = b$. A local trivialization is

$$f \longmapsto (f(\infty), g_{f(\infty)}^{-1} \circ f).$$

Its inverse is $(b, h) \mapsto g_b \circ h$. Postcomposition by a Möbius transformation preserves the degree, and both maps depend holomorphically on their arguments. Since $\mathbb{P}^1$ is a complex homogeneous manifold and hence Oka [25, Proposition 5.5.1], and $\mathcal{R}_d$ is Oka by the preceding argument, the Oka fiber bundle theorem [24, Corollary 1.3], applied to (9.3), gives that R_d is Oka. $\square$

APPENDIX A. THE PLANAR ENLARGEMENT CONSTRUCTION

We follow the two geometric stages of [46, Proposition 2.9]: extending along finitely many radial corridors and filling the intervening sectors. The proof records the parameter domains and chart margins retained at each stage.

Proof of Lemma 2.7. The image of the outer circle has a finite cover by the witness neighborhoods: choose $P_0 \Subset \text{int } \hat{P} \Subset \text{int } P$ and a finite cyclic cover of the outer circle by slightly enlarged arcs $I_1, \dots, I_\ell$ such that there are coordinate witness neighborhoods $V_1, \dots, V_\ell$ with

$$f(I_j \times \hat{P}) \Subset V_j.$$

Extending along finitely many radial corridors. The proof glues in $N = \ell + 1$ operations: one radial gluing followed by one operation for each sector. Choose a chain of parameter domains and positive errors with

$$(A.1) \quad P_0 \Subset \text{int } P^{(N)} \Subset \dots \Subset \text{int } P^{(0)} \Subset \text{int } \hat{P}, \quad \epsilon_k > 0, \quad \sum_{k=1}^N \epsilon_k < \epsilon.$$

The intermediate polydiscs $\widehat{P}$ and $P^{(k)}$ are chosen concentric with P ; we translate the parameter coordinates so that their common center is 0. Choose division points $\zeta_j \in I_j \cap I_{j+1}$, with cyclic indices, so that the arc from ζ_{j-1} to ζ_j lies in I_j and has angular width less than π . Then

$$f(\{\zeta_j\} \times \widehat{P}) \Subset V_j \cap V_{j+1}.$$

By Lemma 2.4, choose a chart spray $f^\sharp$ near $K \times \widehat{P}$; the planar source times a polydisc satisfies its triviality hypothesis. In applying $\mathcal{L}_n$ to this spray, the original parameters $t \in \mathbb{C}^q$ and the fiber variables $v \in \mathbb{C}^n$ together form the parameter tuple $(t, v) \in \mathbb{C}^{q+n}$. Only the planar domain is enlarged. After reducing its fiber radius, choose mutually disjoint thin filled polar rectangles T_j crossing the outer circle at ζ_j so that

$$f^\sharp(T_j \times \widehat{P} \times \overline{\Delta}_\beta^n) \Subset V_j \cap V_{j+1}.$$

Attach to the outer edge of T_j a radial segment L_j ending beyond radius R , and denote its attachment point by s_j^* . Set

$$E_j = T_j \cup L_j, \quad w = (t, v).$$

Let Γ_j be the radial segment from ζ_j to the outer endpoint of L_j . Choose fixed radii $r < a < b$ in the initial portions of the corridors such that

$$\Gamma_j \cap \{a \leq |s| \leq b\} \Subset \text{int } T_j.$$

Fix a radial cutoff on $\{a < |s| < b\}$ and a disc containing all the planar domains used below. Lemma 2.5 then gives a common bound C_* depending only on this fixed data, independently of the subsequent polynomial approximations and narrowings. Choose a target coordinate chart ψ_j whose domain contains V_j , and extend the spray coordinates by

$$u_j(s, w) = \begin{cases} \psi_j(f^\sharp(s, w)), & s \in T_j, \\ \psi_j(f^\sharp(s_j^*, w)), & s \in L_j. \end{cases}$$

The set E_j has connected complement. Write $W = P^{(0)} \times \overline{\Delta}_{\beta/2}^n$. Since the overlap portions lie in the fixed rectangles T_j and the splitting bound is C_* , choose positive numbers η_j^{sp} sufficiently small for the transition estimates, the old-side error ϵ_1 , and the finitely many target-chart margins used below. Uniform continuity of ψ_j^{-1} on a fixed compact coordinate neighborhood then gives $\delta_j > 0$ such that coordinate approximation within δ_j implies target approximation within η_j^{sp} and preserves half of the coordinate-image margin.

Truncate the Taylor expansion in w uniformly on $E_j \times W$ with coordinate error less than $\delta_j/2$. Each of the resulting coefficient functions is continuous on E_j and holomorphic on its interior. Mergelyan's theorem [19, Theorem 3], applied to these finitely many coefficients, therefore gives a polynomial map

$$(A.2) \quad p_j(s, w) = \sum_{|\nu| \leq M} p_{j,\nu}(s) w^\nu, \quad \|u_j - p_j\|_{E_j \times W} < \delta_j.$$

The compact coordinate image has a positive margin inside $\psi_j(V_j \cap V_{j+1})$. Thus, by (A.2), if $\|u_j - p_j\|_{E_j \times W}$ is less than half that margin, then p_j maps $E_j \times W$ into $\psi_j(V_j \cap V_{j+1})$. The same holds on $U_j \times W$ for an open neighborhood $U_j \supset E_j$ thin enough. The map $G_j = \psi_j^{-1} \circ p_j$ is then holomorphic on a neighborhood of $U_j \times W$, maps this product into $V_j \cap V_{j+1}$, and satisfies

$$(A.3) \quad \sup_{T_j \times W} d_Y(G_j, f^\sharp) < \eta_j^{\text{sp}}.$$

Take a bounded open neighborhood O of $K \cup \bigcup_j \Gamma_j$ such that

$$(A.4) \quad \begin{aligned} \overline{O \cap \{a \leq |s| \leq b\}} &\subset \bigcup_j \text{int } T_j, \\ O \cap \{|s| > a\} &\subset \bigcup_j U_j. \end{aligned}$$

Shrink O so that $f^\sharp$ is holomorphic on a neighborhood of $(O \cap \{|s| < b\}) \times W$; the neighborhoods U_j are pairwise disjoint. By (A.4), set

$$A = O \cap \{|s| < b\}, \quad B = O \cap \{|s| > a\}, \quad F_A = f^\sharp, \quad F_B|_{B \cap U_j} = G_j.$$

Consequently

$$\begin{aligned} A \cup B &= O, \quad A \cap B = O \cap \{a < |s| < b\}, \quad K \subset A, \\ \overline{A \setminus B} \cap \overline{B \setminus A} &= \emptyset. \end{aligned}$$

Choose an open polydisc $\tilde{P}^{(1)}$ and radii $\rho_0, \tilde{\rho}_0$ such that

$$P^{(1)} \Subset \tilde{P}^{(1)} \Subset \text{int } P^{(0)}, \quad 0 < \rho_0 < \tilde{\rho}_0 < \beta/2.$$

After shrinking these choices if necessary,

$$\begin{aligned} A \times \tilde{P}^{(1)} \times \Delta_{\rho_0}^n &\subset \text{dom } F_A, \\ (B \cap U_j) \times \tilde{P}^{(1)} \times \Delta_{\rho_0}^n &\subset \text{dom } F_B \quad (1 \leq j \leq \ell). \end{aligned}$$

Apply Lemma 2.5 to (A, B) with passive parameter domain $\tilde{P}^{(1)}$, using the fixed cutoff and the bound C_* . On the overlap and $\tilde{P}^{(1)} \times \Delta_{\rho_0}^n$, write

$$(A.5) \quad \gamma_0(s, t, v) = f^\sharp(s, t, \cdot)^{-1}(F_B(s, t, v)) = v + c_0(s, t, v).$$

Write

$$e_0 = \|c_0\|_{\infty, (A \cap B) \times \tilde{P}^{(1)} \times \Delta_{\rho_0}^n}.$$

Equation (A.3) and Cauchy estimates from the larger fiber polydisc make $c_0, D_v c_0$ in (A.5) satisfy (2.3), with $e = e_0$ and $C_0 = C_*$ by the preceding choice of η_j^{sp} . The same choice ensures that the resulting corrections preserve the target-chart margins on the spoke and sector interfaces. Applying Lemma 2.6 gives corrections a_0, b_0 . For t near $P^{(1)}$, the resulting map and its distance to f satisfy

$$(A.6) \quad \begin{aligned} F_0(s, t) &= \begin{cases} f^\sharp(s, t, a_0(s, t)), & s \in A, \\ G_j(s, t, b_0(s, t)), & s \in B \cap U_j, \end{cases} \\ \sup_{K \times P^{(1)}} d_Y(F_0, f) &\leq \omega_{f^\sharp, K \times P^{(1)}}(C_* e_0) < \epsilon_1. \end{aligned}$$

The two expressions in (A.6) agree on $A \cap B$ by Lemma 2.6. Choose an outer radius $R_* > R$ below the radius of the outer endpoint of every centerline Γ_j ; each centerline therefore meets the circle $\{|s| = R_*\}$. Choose thin polar rectangular neighborhoods U_j^0 of these centerlines with

$$U_j^0 \Subset O \cap U_j, \quad \Gamma_j \cap \{|s| \leq R_*\} \subset U_j^0.$$

They are pairwise disjoint and thin enough that each lies in its own angular wedge. After a cyclic relabeling of the corridors we may also assume that U_{j-1}^0 and U_j^0 bound the sector S_j introduced below. For small corrections their images remain in $V_j \cap V_{j+1}$, and F_0 is holomorphic near the compact set

$$K_0 = K \cup \bigcup_{j=1}^{\ell} \overline{U_j^0}.$$

At this point F_0 is holomorphic near $K_0 \times P^{(1)}$, satisfies the old-side estimate in (A.6), and maps each corridor into the corresponding intersection $V_j \cap V_{j+1}$. These are the inputs for filling the sectors.

Filling the intervening sectors. Let S_j , $1 \leq j \leq \ell$, be the closed polar sector between Γ_{j-1} and Γ_j , of radial range $r \leq |s| \leq R_*$, with indices read cyclically. Write

$$K_j = K_0 \cup S_1 \cup \cdots \cup S_j;$$

then $S_i \cap S_j \subset K_0$ for $i \neq j$, $S_i \cap K_0 = J_i$, and $K_R \subset K_\ell$. Distinct sectors therefore meet only along radial edges contained in K_0 , and the unfilled sectors have the fixed joining sets

$$J_j = K_0 \cap S_j = K_{j-1} \cap S_j.$$

The base arc of J_j lies in I_j and its radial portions lie in $U_{j-1}^0 \cup U_j^0$. By the preceding choice of the approximation errors,

$$F_0(J_j \times P^{(1)}) \Subset V_j \quad (1 \leq j \leq \ell).$$

Claim. Let $1 \leq j \leq \ell$ and write $k = j+1$. Suppose that F_{j-1} is holomorphic near $K_{j-1} \times P^{(k-1)}$ and

$$F_{j-1}(J_i \times P^{(k-1)}) \Subset V_i \quad (j \leq i \leq \ell).$$

Then there is a map F_j holomorphic near $K_j \times P^{(k)}$ such that

$$(A.7) \quad \sup_{K_{j-1} \times P^{(k)}} d_Y(F_j, F_{j-1}) < \epsilon_k$$

and

$$F_j(J_i \times P^{(k)}) \Subset V_i \quad (j < i \leq \ell).$$

Proof of the claim. Choose a thin U-shaped closed band D_j around J_j and a larger disc D'_j such that

$$\begin{aligned} S_j \cap K_{j-1} &\subset \text{int } D_j, & F_{j-1} &\text{ is holomorphic near } D_j \times P^{(k-1)}, \\ S_j &\subset \text{int } D'_j, & D_j &\subset D'_j \text{ is a special disc pair.} \end{aligned}$$

Shrinking the band if necessary, we have

$$F_{j-1}(D_j \times P^{(k-1)}) \Subset V_j.$$

After reducing its fiber radius, a chart spray for F_{j-1} near K_{j-1} therefore satisfies

$$F_{j-1}^\sharp(D_j \times P^{(k-1)} \times \overline{\Delta}_{\beta_j}^n) \Subset V_j.$$

Choose an open band $\mathcal{W}_j \Subset \text{int } D_j$ containing $S_j \cap K_{j-1}$. The compact sets $K_{j-1} \setminus \mathcal{W}_j$ and $S_j \setminus \mathcal{W}_j$ are disjoint. A smooth cutoff separating them, followed by shrinking a bounded neighborhood O_j of K_j , gives bounded planar open sets $A_j, B_j \subset O_j$ such that

$$(A.8) \quad \begin{aligned} K_j \subset O_j &= A_j \cup B_j, & \overline{A_j \setminus B_j} \cap \overline{B_j \setminus A_j} &= \emptyset, \\ A_j \cap B_j &\Subset \text{int } D_j, & B_j \cap K_{j-1} &\Subset \text{int } D_j, \end{aligned}$$

with A_j compactly contained in the base domain of $F_{j-1}^\sharp$ and $B_j \Subset \text{int } D'_j$. Fix the planar cutoff and the resulting splitting bound C_j at this point.

At this operation the n spray variables join the original q passive parameters. Prescribe the retained parameter domains

$$W_j = P^{(k-1)} \times \overline{\Delta}_{\beta_j}^n, \quad W'_j = P^{(k)} \times \overline{\Delta}_{\beta'_j}^n, \quad W'_j \Subset \text{int } W_j, \quad 0 < \rho_j < \tilde{\rho}_j < \beta'_j < \beta_j,$$

before applying $\mathcal{L}_n$. If $j < \ell$, also choose $\mu_j > 0$ such that

$$\{y \in Y : d_Y(y, F_{j-1}(J_i \times P^{(k-1)})) < \mu_j\} \subset V_i \quad (i > j)$$

for every $i > j$; such a common margin exists by the compact containments in the hypothesis of the claim. With C_j , the inverse-chart margins, and this possible remaining chart margin

fixed, choose $\eta_j^{\text{sec}} > 0$ sufficiently small for the gluing estimates below. Property $\mathcal{L}_n$ gives a holomorphic map H_j near $D'_j \times W'_j$ with

$$(A.9) \quad \sup_{D_j \times W'_j} d_Y(H_j, F_{j-1}^\sharp) < \eta_j^{\text{sec}}.$$

After choosing H_j , choose an open polydisc $\tilde{P}^{(k)}$ with

$$P^{(k)} \Subset \tilde{P}^{(k)} \Subset \text{int } P^{(k-1)}.$$

By taking this polydisc sufficiently close to $P^{(k)}$, we ensure

$$(A.10) \quad \begin{aligned} A_j \times \tilde{P}^{(k)} \times \Delta_{\rho_j}^n &\subset \text{dom } F_{j-1}^\sharp, \\ B_j \times \tilde{P}^{(k)} \times \Delta_{\rho_j}^n &\subset \text{dom } H_j. \end{aligned}$$

After further shrinking $\tilde{P}^{(k)}$ about $P^{(k)}$, we also have

$$\sup_{\overline{A_j \cap B_j} \times \tilde{P}^{(k)} \times \Delta_{\rho_j}^n} d_Y(H_j, F_{j-1}^\sharp) < 2\eta_j^{\text{sec}}.$$

Indeed, $\overline{A_j \cap B_j} \Subset \text{int } D_j$ and $\tilde{\rho}_j < \beta'_j$, so this follows from (A.9), continuity, and compactness. Apply Lemma 2.5 to (A_j, B_j) with passive parameter domain $\tilde{P}^{(k)}$. Its splitting bound C_j is unchanged by shrinking this parameter domain.

On the overlap and $\tilde{P}^{(k)} \times \Delta_{\rho_j}^n$, the transition is

$$\gamma_j = (F_{j-1}^\sharp)^{-1} \circ H_j = \text{id} + c_j,$$

where the inverse and composition act in the fiber variable. Uniform continuity of the fixed inverse charts makes c_j uniformly small on the larger fiber product as $\eta_j^{\text{sec}} \rightarrow 0$. Cauchy estimates across the fixed gap $\tilde{\rho}_j - \rho_j$ then make $D_v c_j$ uniformly small on the smaller product. Since C_j is fixed, the smallness conditions (2.3) hold on $\tilde{P}^{(k)} \times \Delta_{\rho_j}^n$ for sufficiently small η_j^{sec} . Write

$$e_j = \|c_j\|_{\infty, (A_j \cap B_j) \times \tilde{P}^{(k)} \times \Delta_{\rho_j}^n}.$$

Applying that lemma gives corrections a_j, b_j such that

$$(A.11) \quad a_j = \gamma_j(s, t, b_j), \quad \|a_j\|_{\infty, A_j \times P^{(k)}} + \|b_j\|_{\infty, B_j \times P^{(k)}} \leq C_j e_j.$$

These corrections give the map near $K_j \times P^{(k)}$

$$(A.12) \quad F_j(s, t) = \begin{cases} F_{j-1}^\sharp(s, t, a_j(s, t)), & s \in A_j, \\ H_j(s, t, b_j(s, t)), & s \in B_j. \end{cases}$$

Lemma 2.6 makes the two expressions agree on $A_j \cap B_j$. On $B_j \cap K_{j-1}$, (A.8) and (A.10) make (A.9) applicable. Hence the continuity modulus ω_j of the fixed old spray gives

$$\sup_{K_{j-1} \times P^{(k)}} d_Y(F_j, F_{j-1}) \leq \eta_j^{\text{sec}} + \omega_j(C_j e_j).$$

By the choice of η_j^{sec} , the preceding upper bound is less than ϵ_k and, when $j < \ell$, also less than $\mu_j/2$. This proves (A.7) and preserves all the remaining chart containments. $\square$

Apply the claim successively for $j = 1, \dots, \ell$, starting with F_0 . Its conclusion supplies the hypothesis for the next sector. Let $F = F_\ell$. Since $K_R \subset K_\ell$ and $P_0 \Subset \text{int } P^{(N)}$, the map F is holomorphic near $K_R \times P_0$. Finally, (A.1), (A.6), and (A.7) give

$$\sup_{K \times P_0} d_Y(F, f) \leq \epsilon_1 + \sum_{k=2}^N \epsilon_k < \epsilon.$$

$\square$

APPENDIX B. THE POLYDISC APPROXIMATION ARGUMENT

Proof of Proposition 2.9. We use the source-coordinate reduction and exhaustion of [46, Proposition 3.4]. We spell out how the thin transverse fibers give the witness control required by Lemma 2.7, and replace the final gluing in that proof by the compositional splitting theorem [21, Theorem 4.1].

Source-coordinate reduction. An affine change of source variables reduces the problem to $Q = \overline{\Delta}_1^m$. Fix $R > 1$ and $\epsilon > 0$. For $m = 1$, compactness gives finitely many witness neighborhoods along the boundary circle, so Lemma 2.7 gives the finite-radius step. We may then pass directly to the exhaustion below. Hence assume $m \geq 2$.

Choose $b > 1$ such that f is holomorphic near $\overline{\Delta}_b^m$. Fix the data of Lemma 2.8 and $\ell > 0$ with

$$1 < r_m < \cdots < r_1 < r_1 + 3\ell < b.$$

Choose $r_1 + 2\ell < b' < b$. On the original polydisc source, Lemma 2.4 gives a chart spray $f^\sharp(x, v)$ near $\overline{\Delta}_{b'}^m \times \overline{\Delta}_\beta^n$, with $f^\sharp(x, 0) = f(x)$. This spray is fixed before the source degree is increased. Fix also $R_+ > R$ and $J \geq 1$ with $a^J R_+ < 1$, and write

$$\Omega = \Delta_{R_+}^m, \quad h_d = H_d^{-J} L^J.$$

Then $h_d(\Omega) \subset E_{d,J}$, and (2.13) gives $h_d \rightarrow \text{id}$ near $\overline{\Delta}_1^m$.

Choose fixed numbers $r_1 < c_0 < c_1 < c_2 < c_3 < r_1 + \ell$ as in (2.10). On the annulus

$$\mathcal{A}_s = \{c_0 \leq |s|^{d^{m-1}} \leq c_3\},$$

the map $\Phi_d(s, \xi)$ lies in $\Delta_{b'}^m$ for large d , and its diameter as ξ ranges over Ξ_2 is $O(1/d)$ by (2.11). Cover the compact image $f(\overline{\Delta}_{b'}^m)$ by finitely many coordinate witness neighborhoods, using a smaller cover with a positive Lebesgue number. Uniform continuity of $f^\sharp$, the thin-fiber estimate, and a reduction of the spray radius ensure that, for each fixed s on $|s|^{d^{m-1}} = c_3$, all values

$$f^\sharp(\Phi_d(s, \xi), v), \quad \xi \in \Xi_2, \quad v \in \overline{\Delta}_\beta^n,$$

lie compactly in one witness neighborhood. The neighborhood may depend on s . Increase d also so that

$$\sup_{\overline{\Delta}_1^m} d_Y(f \circ h_d, f) < \epsilon/2, \quad \sup_{\overline{\Delta}_1^m} |h_d(z)_1| < c_1, \quad r_j(Cc_2)^{1/d} < b' \quad (j \geq 2),$$

where C is the constant in (2.7). Fix this degree and write $h = h_d$; all source geometry and the spray radius now remain fixed.

Apply Lemma 2.7 to $f^\sharp(\Phi_d(s, \xi), v)$ on $\mathcal{A}_s$, with (ξ, v) as passive parameters. Choose a retained transverse polydisc compactly inside Ξ_2 and still containing $\{|\xi_j| \leq B\}$ in its interior, and retain a smaller fiber polydisc with positive margin. The lemma gives a holomorphic map $G(s, \xi, v)$ approximating the whole spray there, with arbitrarily small error, out to a prescribed finite outer radius in s . Choose that radius larger than $c_3^{1/d^{m-1}}$ and larger than

$$\sup\{|s(h(z))| : z \in \Omega, |h(z)_1| \geq c_1\}.$$

This supremum is finite by $|s(h(z))|^{d^{m-1}} = |h(z)_1|$ and boundedness of h on $\overline{\Omega}$; the requirement is vacuous if the set is empty. The inner radius is strictly below $c_1^{1/d^{m-1}}$.

Compositional gluing. On the open cover

$$A = \{z \in \Omega : |h(z)_1| < c_2\}, \quad B' = \{z \in \Omega : |h(z)_1| > c_1\},$$

consider the sprays

$$F_A(z, v) = f^\sharp(h(z), v), \quad F_{B'}(z, v) = G(s(h(z)), \xi(h(z)), v).$$

For $z \in A$, (2.7) gives $|h(z)_j| \leq r_j(Cc_2)^{1/d} < b'$ for $j \geq 2$, while $|h(z)_1| < c_2 < b'$. F_A is therefore the pullback of the chart spray already fixed on the original polydisc; no contractibility of A is needed. Lemma 2.8 and the retained margins give $F_{B'}$ on all of B' . On $A \cap B'$, $\Phi_d(s(h(z)), \xi(h(z))) = h(z)$, so $F_{B'}$ approximates F_A uniformly on a fixed fiber domain.

Fix compact source neighborhoods and concentric fiber balls $\overline{B}_\rho \Subset B_{\rho_+}$ in that domain, with a uniform margin for the inverse fiber charts of F_A . Improving the planar approximation gives the transition

$$\gamma(z, v) = F_A(z, \cdot)^{-1}(F_{B'}(z, v)) = v + c(z, v), \quad \|c\|_\infty \longrightarrow 0,$$

on a fixed neighborhood of the compact overlap used below. The degree, h , Ω , the retained parameter domains, and the inverse-chart margins stay fixed while this error tends to zero.

Choose a compact convex set K and numbers d_1, d_2 such that

$$\overline{\Delta}_R^m \Subset \text{int } K \Subset \Omega, \quad c_1 < d_1 < d_2 < c_2,$$

and set

$$K_A = K \cap \{|h(z)_1| \leq d_2\}, \quad K_B = K \cap \{|h(z)_1| \geq d_1\}.$$

Then $K_A \cup K_B = K$,

$$\overline{K_A \setminus K_B} \cap \overline{K_B \setminus K_A} = \emptyset, \quad K_A \cap K_B \Subset A \cap B'.$$

For the fixed radii $0 < \rho < \rho_+$ above, Cauchy estimates give

$$\sup_{(K_A \cap K_B) \times \overline{B}_\rho} \|D_v c\| \leq C(\rho_+ - \rho)^{-1} \|c\|_\infty \longrightarrow 0.$$

Hence, for sufficiently accurate planar approximation,

$$\Gamma(z, v) = (z, \gamma(z, v)), \quad |\gamma(z, v) - \gamma(z, w)| \geq \frac{1}{2}|v - w|$$

on a fixed smaller fiber ball. After reducing ρ if necessary, Γ is defined and biholomorphic onto its image on an open neighborhood of $(K_A \cap K_B) \times \overline{B}_\rho$. It preserves the vertical foliation $\{z = \text{constant}\}$.

The compact sets $K_A \times \overline{B}_\rho$ and $K_B \times \overline{B}_\rho$ have separated closed fringes, while their union $K \times \overline{B}_\rho$ is convex and hence has a basis of Stein neighborhoods. Thus the ordered pair $(K_B \times \overline{B}_\rho, K_A \times \overline{B}_\rho)$ satisfies the hypotheses of [21, Theorem 4.1]. Let $\omega_{A, \overline{\Delta}_1^m}$ be a continuity modulus of F_A over the unit polydisc. Choose $\eta_{\text{gl}} > 0$ smaller than the reserved fiber margin and such that

$$\omega_{A, \overline{\Delta}_1^m}(\eta_{\text{gl}}) < \epsilon/2.$$

At this point the degree, h , Ω , K_A , K_B , the source neighborhoods, and the fiber margins are fixed. The inverse-chart bounds, the Cauchy bounds, and the smallness threshold in the splitting theorem therefore remain fixed as the planar approximation error tends to zero. Making $\|c\|_\infty$ sufficiently small, the foliation-preserving form of the splitting theorem gives neighborhoods $U_A \supset K_A$, $U_B \supset K_B$ and fiber maps α_A, α_B satisfying

$$\gamma(z, \alpha_B(z, v)) = \alpha_A(z, v), \quad \|\alpha_A - \text{id}\|_\infty + \|\alpha_B - \text{id}\|_\infty < \eta_{\text{gl}}.$$

After restricting $U_A \Subset A$ and $U_B \Subset B'$, the map

$$\tilde{f}(z) = \begin{cases} F_A(z, \alpha_A(z, 0)), & z \in U_A, \\ F_{B'}(z, \alpha_B(z, 0)), & z \in U_B \end{cases}$$

is holomorphic near $\overline{\Delta}_R^m$. Since $\overline{\Delta}_1^m \subset K_A$, we have

$$\sup_{\overline{\Delta}_1^m} d_Y(\tilde{f}, f) \leq \omega_{A, \overline{\Delta}_1^m}(\eta_{\text{gl}}) + \sup_{\overline{\Delta}_1^m} d_Y(f \circ h, f) < \epsilon.$$

Exhaustion. The preceding construction proves extension to every prescribed finite radius for all $m \geq 1$. Set $f_0 = f$. As in the final exhaustion in the proof of [46, Proposition 3.4],

choose radii $1 = R_0 < R_1 < \cdots \rightarrow \infty$ and, after an affine rescaling at each stage, maps f_j holomorphic near $\bar{\Delta}_{R_j}^m$ such that, for $j \geq 1$,

$$\sup_{\bar{\Delta}_{R_{j-1}}^m} d_Y(f_j, f_{j-1}) < \epsilon 2^{-j-1}.$$

Completeness of d_Y gives a limit F , uniformly on compact subsets of $\mathbb{C}^m$. Moreover, $\sup_{\bar{\Delta}_1^m} d_Y(F, f) < \epsilon$. In a target chart, local uniform convergence and the scalar uniform limit theorem give $F \in \mathcal{O}(\mathbb{C}^m, Y)$. $\square$

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STATEMENT ON THE USE OF ARTIFICIAL INTELLIGENCE

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